

AN ANALYSIS OF LYME DISEASE DATA WITH MACHINE LEARNING TOOLS

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ABSTRACT. We present an exploration of Lyme disease data using diverse machine learning techniques, including Non-negative Matrix Factorization (NMF), classification algorithms, and anomaly detection. Our study focuses on a survey data set collected from individuals who have been affected by Lyme disease. Leveraging NMF, we uncovered underlying patterns that contribute to a better understanding of the disease's symptoms, including indicators that lead to neurological manifestations of the disease. In addition, by employing classification techniques such as logistic regression and SVM, we achieve accurate differentiation between chronic and early lyme individuals, as well as accurately predict whether someone was diagnosed before or after a year prior to onset of symptoms. We also leveraged anomaly detection methods to uncover patients with symptoms that may lead to extreme, or uncommon symptoms and manifestations, which we then analyzed with semi-supervised NMF. We developed two helper functions based on feature similarity to improve upon Neural NMF and a multiplex community detection algorithm implemented on the patient network with decent interpretability. Moreover, we propose a Fair NMF algorithm where different subgroups converge on lower rank and result in fairer low dimensional representation. Through our various mathematical methods, our study enhances our knowledge of Lyme disease and various mathematical techniques.

1. INTRODUCTION

Lyme disease is the most common vector-borne disease in the United States, with approximate 476,000 Americans who are diagnosed and treated for Lyme disease a year [CCS⁺23]. However, this number is an underestimate of the actual number of people impacted by Lyme Disease a year because there is not currently a clear way to diagnose Lyme Disease, nor do all cases of Lyme disease get reported. The diagnosis of Lyme Disease is difficult, as the symptoms of Lyme Disease are quite similar to that of other chronic diseases. The cost of current tests is expensive, rather timely, and can yield inaccurate results [HCM⁺14]. Due to these concerns, there is also confusion and controversy surrounding the chronic form of Lyme Disease. Like in other applications, there is a vast amount of data potentially available, but without innovative mathematical techniques to optimally extract crucial information, scientific progress will remain stagnant. The traditional research and treatment models are extremely costly, slow, and often ineffective. The controversy, stagnation, high cost, poor treatment, and inaccurate diagnosis of Lyme disease make it the perfect motivating example for our project.

Questions of interest, that perhaps could be addressed by this study include (i) how to more accurately and quickly diagnosis Lyme (note that prolonged diagnosis seems to correlate with the chronic form of the illness and many patients do not get diagnoses for years after infection), (ii) how to better treat the illness (there is currently no consensus on what types of treatment [which antibiotics, supplements, alternative treatment protocols] work best, and perhaps this differs for different patient types, (iii) why the chronic form of the illness appears to be more common in females, (iv) what kinds of co-infections are correlated with particular symptoms or geographic regions, and how those co-infections are best treated, (v) what are the best predictors for a patient feeling well or going into remission, (vi) what describes a neurologic mainfestion and are there predictors or treatment factors, (vii) comparing well and unwell

groups, (viii) comparing acute versus persistent patients, (ix) time to diagnosis and duration of illness, (x) how to determine whether a patient as neurological or not.

Given the difficulties patients have experienced in recovering from Lyme disease, we aim to discover underlying patterns among well and unwell patients using Nonnegative Matrix Factorization (NMF) as a tool. Compared with Principal Components Analysis (PCA) and Vector Quantization (VQ), NMF is a parts-based learning algorithm and requires nonnegativity constraints[LS99]. It helps to reduce the dimensionality of our data set and reveals significant information about patients' features concerning their particular situations. One of our main focuses is extracting symptom patterns in Lyme patients with unsupervised NMF to illustrate what leads to the different statuses among well and unwell patients. Moreover, we implement Semi-supervised NMF (SSNMF) as a classification tool to study which features contribute to neurological manifestations for chronically unwell patients. In this paper, we present our findings in studying Lyme patients with the basic use of unsupervised NMF for feature extraction and SSNMF for classification.

In light of the increasing concerns regarding biases and discriminatory outcomes stemming from traditional dimensionality reduction techniques, we focus our attention on the development of fair dimensionality reduction methods. NMF is heavily used in fields such as image processing [LKL14], text mining [ZEAC13], or bioinformatics [LJZ08]. It can also be used as a dimensionality reduction technique. [TSKK01]. These methods are designed to counteract biases present in real-world datasets, which could stem from attributes like gender, race, or ethnicity. Our interest is in mitigating these biases and ensuring that the resulting data representations remain uninfluenced by sensitive attributes. Principal Component Analysis (PCA), a traditional dimension reduction method, projects the original dataset into a lower-dimensional space by linearly combining the data variables. However, such techniques can inadvertently exacerbate biases. This observation persists even when both populations are equally represented in the dataset. To address this, we propose a new concept called Fair NMF, building upon the existing Fair PCA algorithm [STM⁺18]. Our approach encompasses various methodologies, including rescaling data to examine the loss of subgroups and introducing a Weighted-Average Dictionary Matrix to stabilize the update process. Furthermore, we present an early stopping strategy that halts the algorithm when subgroup losses converge within a certain tolerance, thereby ensuring fairness and convergence. This paper details these approaches, providing a comprehensive understanding of Fair NMF's mechanisms and its contributions to promoting equitable dimensionality reduction. In this paper, we will also further explore and formulate methods for fair dimension reduction using NMF and test it using both synthetic and Lyme data.

1.1. Contributions. In our study, we made significant contributions to the understanding and analysis of Lyme disease patient data. Using unsupervised Non-Negative Matrix Factorization (NMF), we successfully identified underlying symptom patterns in Lyme patients, shedding light on potential commonalities among cases. Furthermore, through supervised NMF, we predicted neurological manifestations in chronically unwell patients, enhancing our ability to anticipate and address severe symptoms.

Our work also extended to anomaly detection. Leveraging anomaly detection algorithms, namely Isolation Forest (iForest) and Local Outlier Factor (LOF), we effectively identified patients exhibiting unusual symptom profiles. By subsequently employing supervised NMF, we uncovered concealed patterns that significantly contribute to determining an anomaly.

In addition, we pioneered the development of a patient network that responds to NMF-based multiplex community detection, offering a comprehensive view of patient interactions and shared characteristics. To enhance the Neural NMF approach, we introduced two novel helper functions rooted in feature similarity, resulting in improved model performance.

Additionally, we formulated a fair NMF approach, which we rigorously tested on both synthetic data and real-world MyLymeData. This formulation addresses the challenges of subgroup variations and promotes equitable low-dimensional representations. Our findings revealed a non-linear relationship between loss and scalars, prompting the design of the Fair NMF algorithm. In contrast to vanilla NMF and the use of early stopping, the proposed Fair NMF algorithm showcased superior convergence in subgroups while maintaining lower ranks.

Overall, our research contributes a comprehensive suite of methodologies for analyzing Lyme disease patient data, offering deeper insights into symptom patterns, predictions, anomaly detection, and equitable representations.

1.2. Organization. The structure of the paper unfolds as follows. We start in Section 2 by presenting an overview of the MyLymeData dataset employed throughout the study, accompanied by an introduction to Lyme Disease organization involved. Moving on to Section 3, we provide essential background information and discuss pertinent previous works, setting the stage for introducing the methodologies and models deployed in our experiments. A comprehensive exploration of the adopted methods, complemented by their corresponding pseudocodes, is undertaken in Section 4. We then proceed to subject the developed algorithms to synthetic datasets in Section 5. In Section 6, we elucidate the preprocessing procedures applied to the Lyme data. Finally, Section 7 delves into the MyLymeData experiments, offering a comprehensive breakdown of the undertaken analyses and outcomes.

2. LYME DISEASE DATA

In Lyme’s early, or acute form, the disease may cause a hallmark erythema migrans (EM) rash and/or flu-like symptoms such as fever, malaise, fatigue, and generalized achiness [AMM⁺09]. Although most cases recover after 2-4 weeks of antibiotics, it is estimated that as many as 36% of those diagnosed and treated early remain ill after treatment [ARCK13].

Post-treatment Lyme Disease (PTLD) or Chronic Lyme disease (CLD) describes the condition in which patients experience symptoms post-treatment of Lyme disease. The most frequently reported three worst symptoms in patients are neurologic-associated symptoms (84%), fatigue (62%), and musculoskeletal-associated symptoms (57%) [JSM18]. Although doctors agree that the acute form of Lyme disease exists, there is controversy over the existence of the chronic form. This has led to the creation of two sides, “mainstream” and “Lyme-literate” [Dum20]. Those who support the “mainstream” side argue that the standard of care in regard to the diagnosis and treatment of Lyme disease is straightforward. On the other side, those who support the “Lyme-literate,” state that Lyme disease tests are often inaccurate, and Lyme disease can lead to “chronic Lyme disease” [Dum20].

According to data collected from our collaborators at LymeDisease.org, 72% of Lyme disease patients report they were initially misdiagnosed [JSM18]. With more than half of patients (51%) report that it took them more than three years to be diagnosed and roughly the same proportion (54%) saw five or more clinicians before diagnosis [JSM18]. This is problematic, as evidence has also shown that prolonged diagnosis leads to a higher risk of developing the chronic form of Lyme Disease.

However, despite the high incidence and severity of Lyme disease, shockingly little research has been done, both clinically and analytically [KHE⁺01, KHG⁺03, FKC⁺08, DBMP12, KHE⁺01].

2.1. LymeDisease.org. Established in 1989, LymeDisease.org (LDo) is one of the oldest and most trusted Lyme disease organizations in the country dedicated to improving patient care, advancing Lyme disease research, and impacting healthcare policy. Grounded in the principles of patient empowerment, it is the largest communication network representing Lyme disease patients in the nation with widely distributed website content, blogs, and social media. It supports science-based research and drives legislative change to amplify the collective voice of Lyme patients and increase public awareness about Lyme disease. It is also a trusted and knowledgeable intermediary responsible for advancing the interests of

the Lyme community and curating the use of patient data to promote the mission and values of Lyme patients[lym].

2.2. MyLymeData. In November 2015, LDo announced the launch of MyLymeData, a patient-powered research project [Wor13]. The goal of patient-powered registries is to ensure that the research agenda moving forward is managed and controlled by patients. MyLymeData is the first national large-scale study of both acute Lyme disease and chronic Lyme disease. By giving the opportunity to patients to take charge, patients will be able to improve their outcomes. Patients have access to their electronic health records, insurance records, and lab results/biorepositories. As well as a deep understanding of their symptoms and symptom response. Today, MyLymeData is in the top 10% of patient-powered research registries in the country. It has enrolled over 17,000 patients nationwide in the registry to date and continues to grow.

MyLymeData collects information from patients about diagnosis, demographics, quality of life, functional impairment, treatment, and symptoms. A critical feature of the patient-generated data is survey branching, which means that the next question respondents answer is determined by their previous responses.

3. BACKGROUND AND RELATED WORK

3.1. Notation & Terminology. In this section, we introduce some notation. We denote a matrix in boldface $\mathbf{X}$ and the matrix entries with a_{ij} .

$\mathbf{A} \geq 0$	A nonnegative matrix $\mathbf{A}$
$\mathbb{R}_+^{m \times n}$	Set of all real nonnegative matrices of size $m \times n$
$\mathbf{A} \circ \mathbf{B}$	Hadamard product of $\mathbf{A}$ and $\mathbf{B}$
$\ \cdot\ _1$	1-norm $\ \mathbf{A}\ _1 = \max_{1 \leq j \leq n} \sum_{i=1}^m a_{ij} $
$\ \cdot\ _F$	Frobenius norm $\ \mathbf{A}\ _F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n a_{ij} ^2}$
$\mathcal{D}[i]$	Value stored at index i of dictionary $\mathcal{D}$

When we write rank, we are referring to nonnegative rank unless specified.

3.2. Nonnegative matrix factorization (NMF). In [LS99], the authors demonstrated nonnegative matrix factorization (NMF) as a parts-based learning algorithm and compared NMF with Principal Components Analysis (PCA) and Vector Quantization (VQ). Given a nonnegative matrix $\mathbf{X} \in \mathbb{R}_+^{n_1 \times n_2}$ with columns representing samples and rows representing features, and a target dimension $r \in \mathbb{N}$, NMF seeks to decompose $\mathbf{X}$ into a product of two low-dimensional nonnegative matrices $\mathbf{A}$ and $\mathbf{S}$ which satisfy the following

$$\min_{\mathbf{A} \in \mathbb{R}_+^{n_1 \times r}, \mathbf{S} \in \mathbb{R}_+^{r \times n_2}} \|\mathbf{X} - \mathbf{AS}\|_F^2$$

Here, $\mathbf{A}$ is called the dictionary matrix, and $\mathbf{S}$ is called the representation matrix. Typically, r is chosen such that $r < \min\{n_1, n_2\}$ to reduce the dimension of the original data matrix or reveal latent themes in the data.

NMF requires nonnegativity constraints that only allow nonnegative additive combinations which, for example, enables approximations of an image by nonnegative additions of parts or basis images [LS99]. It is also shown to perform well in the semantic analysis of text documents. Two standard methods for determining nonnegative factors $\mathbf{A}$ and $\mathbf{S}$ were studied in [LS00]. Multiplicative update rules are introduced for implementing these two methods by preserving their non-increasing property and comparing them with the additive update rules derived from gradient descent.

In the case of studying Lyme patients' data, suppose we have a nonnegative data matrix $\mathbf{X}$ with rows representing questions and columns representing patients. Therefore, each column of $\mathbf{A}$ represents how

important each question is for each topic and each column of $\mathbf{S}$ represents how important each topic is for each patient, and the following energy function is minimized using the Forbenius norm.

$$E(\mathbf{A}, \mathbf{S}) = \frac{1}{2} \|\mathbf{X} - \mathbf{AS}\|_F^2. \quad (3.1)$$

Matrix $\mathbf{A}$ helps us to understand how questions contribute to topics, whereas matrix $\mathbf{S}$ helps us to interpret how patients cluster by topics. In this paper, we first implement the standard NMF to discover underlying patterns around different subgroups of Lyme patients for symptoms. We also implement various versions of unsupervised and supervised NMF to discover interesting trends by studying MyLy-meData.

3.3. Semi-supervised NMF. Semi-supervised NMF (SSNMF) jointly incorporates the data matrix and the (partial) class label matrix into NMF, sharing a common factor matrix for consistency [LYC09]. The sum of weighted residuals is minimized using multiplicative updates in the decomposition of the data matrix and the label matrix to handle missing data and to incorporate partially labeled data. This method improves clustering and classification performance, compared to the standard NMF. The optimization problem of SSNMF is formulated as follows [LYC09]:

$$\underset{\mathbf{A}, \mathbf{S}, \mathbf{B} \geq 0}{\operatorname{argmin}} \|\mathbf{W} \odot (\mathbf{X} - \mathbf{AS})\|_F^2 + \lambda \|\mathbf{L} \odot (\mathbf{Y} - \mathbf{BS})\|_F^2. \quad (3.2)$$

where $\mathbf{W}$ is a binary weight matrix that indicates missing data with 0 for unobserved data, and 1 for observed data; similarly, the weight matrix $\mathbf{L}$ handles missing labels with 0 for the absence of a label and 1 otherwise; $\mathbf{S}$ is a shared representation matrix that provides coefficients for each sample data; and $\mathbf{B}$ provides how labels and topics are related.

In [HKL⁺20], a classification framework is proposed using SSNMF with the following three steps:

- (1) Run SSNMF using the train set to learn the dictionary matrices $\mathbf{A}$ and $\mathbf{B}$.
- (2) Find the representation matrix of the test set, $\mathbf{S}_{test}$, by applying the non-negative least square using the test set data matrix.
- (3) Predict labels of the test set using the matrix product $\mathbf{BS}_{test}$.

3.4. Neural NMF. Deep Learning is an exploding area of machine learning based on data representations using multiple levels of abstraction. Though Deep Learning gained social attention and awareness because of its ability to process huge amounts of data, its implications pose problems, including difficulty in interpreting and understanding mathematically. To gain better interoperability for Deep Learning, Non-Negative Matrix Factorization (NMF), a dimension reduction technique easy to formulate and interpret, was artfully combined with Neural Networks to produce reliable and explainable results. In [FH17], a deep NMF model is proposed consisting of multilayered NMF combined with a pooling layer followed by backpropagation. NMF is used as a substitute for convolution and after each NMF operation, a pooling function is used on the resulting matrix. The process is then repeated until the very last layer, where semi-supervised NMF is used to create a label learning energy function. Semi-supervised NMF is used as a substitute for the fully connected multi-layer perceptron in deep learning. Then the model is trained using backpropagation. The paper [GHM⁺19] was one of the most recent in the field on combining Neural Network structures and NMF as a variant of the original Deep NMF paper [CZ⁺06] which uses back propagation via multiplicative rules to determine each step. Neural NMF implements gradient descent with backpropagation to iterate to the next step.

3.5. Topic Modeling Mixed with Network Structure. An advanced topic model with network structure embedded is capable of enriching the interpretability of standard models with an integrated inference generalizing different types of relationships between data points. In addition to the routine task of identifying latent topics and document-word distributions, such a model can further discover the intrinsic connections between individual nodes and groups of nodes, making the data mixture more structurally pronounced and organized. Studies in the field have laid the foundation for the framework of numerous

variations of such models. As one major form of the topic models, NMF bears the clustering capability and can also be used as the framework for methods like community detection, which is introduced in more detail in the next section. Some related works in community detection using NMF include a general survey [HFC⁺22] synthesizing the most up-to-date efforts done on various forms of network, a paper [GPZ19] delineating a framework of NMF-based community extraction, etc.

3.5.1. Community Detection. Community detection methods refer to algorithms able to locate communities in a complex network structure. The identified communities are commonly groups of nodes closely connected by edges representing certain types of relationships. It is widely adopted in network analysis for both single layer model and multiplex structure like social network, in which connections are stratified into layers indicating different types of relationships, including friends, family, colleagues, etc. The AUCS dataset introduced in [KL15] follows precisely such multiplex structure. An example multiplex network model with some established communities is shown in Figure 1, where the nodes joined differently on each layer share an integrated pattern in favor of grouping the communities across layers.

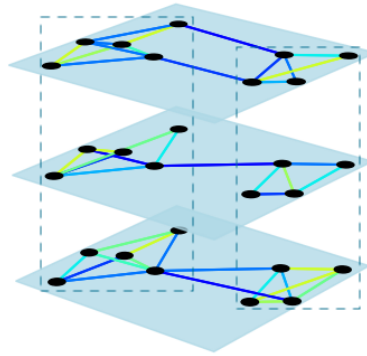


Figure 1 This is an example multiplex network generated using Pymnet library with 8 nodes on each of the 3 layers connected by weighted edges. Two communities enclosed by dotted boxes can be identified across the different layers.

3.5.2. Nonnegative Matrix Tri-Factorization. The community detection framework which we will look into in this paper involves nonnegative matrix tri-factorization (NMTF). NMTF is an extension of NMF that allows more degrees of freedom [PSCL20]. Unlike NMF, NMTF sets two rank parameters to reduce dimensionality. It aims to find an approximation of $\mathbf{X} \in \mathbb{R}_+^{m \times n}$ in the form of 3.3, where $\mathbf{U} \in \mathbb{R}_+^{m \times l}$, $\mathbf{S} \in \mathbb{R}_+^{l \times k}$, and $\mathbf{V}^T \in \mathbb{R}_+^{k \times n}$.

$$\mathbf{X} \approx \mathbf{USV}^T \quad (3.3)$$

3.6. Anomaly Detection. Anomalies are classified as data points that stand out amongst other data points in the data set i.e., deviate from the data set’s normal behavioral patterns. Anomalies in the health field can be due to the unusual health condition or activity of a patient. They also detect false information diluting the data pool. However, detecting anomalies is a difficult task as in different contexts, the definition of anomaly can vary.

Isolation forests (iForest) is an unsupervised algorithm that focuses on the principle that anomalies are few and different, and thus more susceptible to isolation [LTZ08]. By randomly selecting both a feature from the dataset and a split point (called an isolation score) between the minimum and maximum values of the selected attribute in the sample partitions of the data are created, creating a collection of binary decision trees. This process is repeated recursively until each data point is isolated in its own trees. Anomalies tend to be isolated more quickly since they require fewer splits to separate them from normal instances, meaning that anomalies would be the points with the shortest average path length. A

threshold is typically set to determine which instances are considered anomalies based on their average path lengths. Instances with average path lengths below the threshold would be identified as anomalies.

$$s(x, T) = 2^{-\frac{E(h(x))}{c(n)}} \quad (3.4)$$

where,

$s(x, T)$ is the anomaly score of instance x in tree T .

$h(x)$ is the path length of instance x in tree T , i.e., the depth of x in the tree.

$c(n)$ is a normalization factor that depends on the average path length of unsuccessful searches in a Binary Search Tree (BST) with n nodes.

For n instances, $c(n)$ can be approximated as $2 \cdot \left(H(n-1) - \frac{n-1}{n} \right)$,

where $H(n-1)$ is the $(n-1)$ -th harmonic number.

The anomaly score $s(x, T)$ is calculated for each tree in the forest, and a lower score indicates a higher likelihood of being an anomaly.

Local Outlier Factor (LOF) is an unsupervised density-based algorithm that relies on the idea that anomalies are typically grouped away from the main group, thus having a lower density. This is calculated by computing the local density deviation with respect to its neighbors, this number is called the LOF score. If the LOF score is greater than the threshold set, then it is considered an outlier [BKNS00].

The equation for calculating the Local Outlier Factor for a data point x can be represented as follows:

$$LOF(x) = \frac{\sum_{o \in N_k(x)} \frac{\text{reach-dist}(x, o)}{\text{reach-dist}_k(o)}}{k \cdot \sum_{o \in N_k(x)} \frac{\text{reach-dist}(x, o)}{\text{reach-dist}_k(o)}} \quad (3.5)$$

where,

$LOF(x)$ is the Local Outlier Factor of data point x .

$N_k(x)$ is the set of k nearest neighbors of data point x .

$\text{reach-dist}(x, o)$ is the reachability of the distance between data points x and o .

$\text{reach-dist}_k(o)$ is the reachability distance of the data point o to its k -th nearest neighbor.

k is the number of neighbors considered (a user-defined parameter).

A higher value of $LOF(x)$ indicates that data point x is more likely to be an outlier in the dataset. We use SSNMF in our experiments to interpret the results of the two anomaly detection algorithms above (e.g. see Section 3.3).

3.7. Fairness in Machine Learning. When data used to train algorithms is biased, the resulting model will reflect these biases. Sampling bias and measurement bias can create disparity in the model [MMS⁺21]. Additionally, any biases in the algorithm itself can create discrepancies. In medical applications specifically, when the data is skewed towards certain populations, this can result in misclassification and dangerous consequences for the underrepresented populations. Methods to mitigate the biases in machine learning algorithms and achieve fairness typically fall under either pre-processing, in-processing, or post-processing algorithms. Pre-processing attempts to remove bias from the data prior to running the algorithm. [CWV⁺17] In-processing attempts to manipulate the algorithm to remove biases. [ZVRG17] This can be achieved through implementing constraints or adjusting the objective function. Finally, post-processing can be done when results from a biased model are reassigned in an attempt to diminish the bias. [HPS16] There are also protected attributes that, by law, should not affect the outcome of a decision in housing or credit card decision-making. Some attributes, such as race, color, marital status, and age. [MMS⁺21].

3.7.1. *Fair Classification.* Classification is a machine learning task that involves assigning input data points to predefined categories or classes based on their inherent characteristics. Classification is driven by the desire to create a decision boundary that effectively separates data instances into distinct groups, allowing the model to generalize its understanding and make accurate predictions on unseen data.

The performance of classifiers is negatively affected if an imbalanced dataset is used in training. This happens in the real world when the minority is statistically underrepresented during data collection. The previous effort done to address the issue includes under-sampling the majority and re-sampling the minority in varying fashion (either re-sample the entire class or only boundary data), while the author of the paper [CBHK02] designs a combination of the approaches. Synthetic Minority Oversampling Technique (SMOTE) is an algorithm that takes in three parameters (minority sample size, amount of SMOTE $N\%$, and number of nearest neighbor) and returns a synthesized sample of size $N\%$ of the original minority class. Synthesized examples are generated from the line segments joining randomly chosen nearest neighbors of the input data. SMOTE has improved accuracy compared to the plain implementation of under-sampling.

Multitask learning frameworks focus on increasing the accuracy of each group, rather than maximizing the accuracy of the whole sample. Preferential sampling is also used to create a discrimination-free training dataset [MMS⁺21]

Demographic parity is a fairness criterion that aims to ensure that the predictions of a machine learning model are independent of the protected attribute [MMS⁺21] [VR18], where the model’s predictions should be equally accurate across different groups defined by the protected attribute. The demographic parity score in the Fairlearn package [Hil23] in python measures the difference in prediction rates between different groups. It is calculated as the absolute difference between the prediction rate for the privileged group and the prediction rate for the unprivileged group. A lower demographic parity score indicates a fairer model, as it means the difference in prediction rates between the groups is reduced. Other metrics also exist depending on the ML task, such as equalized odds and false negative rates.

3.7.2. *Fair Dimension Reduction.* Many real-world datasets contain biases and discrimination based on protected attributes like gender, race, or ethnicity. Traditional dimensionality reduction techniques may inadvertently amplify these biases, leading to unfair or discriminatory outcomes, which motivates us to develop fair dimensionality reduction methods that aim to mitigate such biases and ensure that the resulting representations are not influenced by sensitive attributes.

One of the traditional dimension reduction methods is Principal Component Analysis (PCA) where the original dataset is projected onto a lower dimensional space, with each principal component being the linear combination of the data variables. Fair dimension reduction is an active research area. For instance, [OA19] and [LKO⁺22] are two papers that offer formulations of fair PCA. The first paper states that fairness is achieved if information about protected attributes cannot be inferred from the resulted dimensionality-reduced points, and the latter incorporates fairness in PCA by minimizing the maximum mean discrepancy of dimensionality-reduced conditional distributions of protected attributes.

Another leading paper on the topic of fair PCA is [STM⁺18], where the authors focus on assessing whether PCA inadvertently generates data representations that exhibit unfair results between two distinct populations. By examining real-world datasets, it is demonstrated that PCA produces higher reconstruction errors for different populations. This discrepancy persists even when both populations are equally represented in the dataset, such as in cases examining people with higher and lower levels of education or different sexes. These findings serve as the impetus for the exploration of dimension reduction methods that ensure comparable accuracy for both populations. The paper introduces a new notion of Fair PCA and presents a polynomial-time algorithm capable of identifying a low-dimensional data representation that achieves near-optimal results based on the concept of reconstruction loss. Important notions are introduced in the paper:

Definition 3.1. (Reconstruction Loss) Given a matrix $\mathbf{Y} \in \mathbb{R}^{a \times n}$, let $\hat{\mathbf{Y}} \in \mathbb{R}^{a \times n}$ be the optimal rank- d approximation of $\mathbf{Y}$. For a matrix $\mathbf{Y} \in \mathbb{R}^{a \times n}$ with a rank at most d we define

$$\text{loss}(\mathbf{Y}, \mathbf{Z}) := \|\mathbf{Y} - \mathbf{Z}\|_F^2 - \|\mathbf{Y} - \hat{\mathbf{Y}}\|_F^2$$

Then, the notion of fairness is achieved when the maximum loss suffered by any group is minimized.

Fair PCA is formulated in [STM⁺18] as follows. Given m data points in $\mathbb{R}^n$ with subgroups A and B , we define the problem of finding a fair PCA projection into d -dimensions as,

$$\min_{\mathbf{U} \in \mathbb{R}^{m \times n}, \text{rank}(\mathbf{U}) \leq d} \max \left\{ \frac{1}{|A|} \text{loss}(\mathbf{A}, \mathbf{U}_A), \frac{1}{|B|} \text{loss}(\mathbf{A}, \mathbf{U}_B) \right\}$$

where $|A|$ indicates the size of population A . This fair PCA problem is further relaxed to a semidefinite optimization problem. Lastly, the paper validates the effectiveness of the fair PCA algorithm by generating a fair and concise low-dimensional representation of the data in various real-world datasets.

4. METHODS

4.1. Nonnegative Matrix Factorization (NMF). In our experiments, we used the built-in function for NMF from the python package scikit-learn [PVG⁺11]. By extracting factored matrices from the decomposition, we studied the topic representations among different groups by plotting a heatmap for selected features.

4.2. Semi-supervised NMF (SSNMF). For the purposes of our study, we used the python package called ssnmf [HKL⁺20], which also includes several models for SSNMF. Each model also consists of a label matrix, $\mathbf{Y}$, classification factor matrix, $\mathbf{B}$, and classification weight parameter, λ . However, as opposed to other variations of NMF, this model generalizes NMF to supervised learning tasks and provides a topic model which simultaneously provides a lower dimensional representation of the data and a predictive model for targets.

To use SSNMF for classification, we split the data set into training, validation, and testing sets. We first use the training and validation set to search for the “optimal” λ that gives the best F1 score in the classification task by testing with different values for λ . Then, we combine the training and validation set to train our model with the optimal λ and apply it to the test set. We compute the product of matrices $\mathbf{B}$ and $\mathbf{S}$ from the trained model and for the binary case, we assign the entry of each entry greater than 0.5 to 1 and all rest to 0. We plotted the heatmap using the obtained matrix to represent how each feature contributes to the studied label.

4.3. Neural NMF Improvements. Implementation of Neural NMF met certain obstacles. When we ran one layer Neural NMF with eight groups on a strictly tree-structured, three-layered synthetic data frame, some groups resulted in zero members. Certain groups were clustered together, resulting in zero members in others. For instance, in a synthetic data experiment demonstrated in section 5, the 200 data points were grouped into groups of 25, 25, 50, 25, 25, 25, 25, and 0. One group contained no actual data points, indicating that the hard split was not through. Instead, two groups clustered together, creating a group that is at a higher level in the data structure with respect to the other data points.

To detect and deal with these clusters of groups, we propose an algorithm that accurately captures these groups and cleanly identifies them. Our input is a list of groups in the form of matrices of shape $c \times n$, where c is the group size, and n is the fixed number of features. We use cosine similarity as an indicator of feature similarity. Usually, cosine similarity is defined between two matrices $\mathbf{A}$ and $\mathbf{B}$ with the same dimensions, and calculated by

$$\cos(\mathbf{A}, \mathbf{B}) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum_{i=1}^n \mathbf{A}_i \mathbf{B}_i}{\sqrt{\sum_{i=1}^n (\mathbf{A}_i)^2} \sqrt{\sum_{i=1}^n (\mathbf{B}_i)^2}} \quad (4.1)$$

We propose a novel way to calculate the cosine similarity between two matrices of the same number of rows but different number of columns. Given matrix $\mathbf{A}$ of size $a \times n$ and a matrix $\mathbf{B}$ of size $b \times n$, we

calculate the row-wise cosine similarity, i.e., given matrix $\mathbf{A} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_a \end{bmatrix}$ and matrix $\mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_b \end{bmatrix}$, we calculate

$C_{ij} = \cos(a_i, b_j)$ for all combinations of i and j and create a matrix whose ij th value corresponds to the value of C_{ij} . We call this the cosine similarity matrix, which can be expressed mathematically as

$$\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1b} \\ \vdots & \vdots & \cdots & \vdots \\ C_{a1} & C_{a2} & \cdots & C_{ab} \end{bmatrix}.$$

After normalizing the cosine similarity matrix row-wise by dividing each element by the sum of every element in the row, we define the cosine value as the Frobenius norm of the normalized cosine similarity matrix. Then, we sum up the Frobenius norm of each group with respect to other groups and form a list of summations of different groups. With a given quantile level, we claim that the groups with summations equal to or above the quantile threshold in the population need further splitting.

A detailed pseudo-code for this process is given below in the form of an algorithm.

Algorithm 1: Feature Similarity-Based Cluster Detection

Input: A quantile level α .

A list L of m groups in the form of matrices of shape $c_m \times n$ (c the group size, n the fixed number of features) for cluster detection. At least one of the groups contains zero data points.

Result: A list of groups that needs further splitting.

for $i \in \{1, \dots, m\}$ **do**

for $j \in \{i + 1, \dots, m\}$ **do**

 Given i th matrix $\mathbf{A} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_a \end{bmatrix}$ and j th matrix $\mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_b \end{bmatrix}$, calculate $C_{pq} = \cos(a_p, b_q)$ for all

 combinations of p and q and obtain the cosine similarity matrix $\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1b} \\ \vdots & \vdots & \cdots & \vdots \\ C_{a1} & C_{a2} & \cdots & C_{ab} \end{bmatrix}$.

 Obtain normalized cosine similarity matrix $\begin{bmatrix} \frac{C_{11}}{\sum_{r=1}^b C_{1r}} & \frac{C_{12}}{\sum_{r=1}^b C_{1r}} & \cdots & \frac{C_{1b}}{\sum_{r=1}^b C_{1r}} \\ \vdots & \vdots & \cdots & \vdots \\ \frac{C_{a1}}{\sum_{r=1}^b C_{ar}} & \frac{C_{a2}}{\sum_{r=1}^b C_{ar}} & \cdots & \frac{C_{ab}}{\sum_{r=1}^b C_{ar}} \end{bmatrix}$.

 Calculate the Frobenius norm of the normalized cosine similarity matrix.

end

 Sum up the Frobenius norm of each group with respect to other groups and form a list of summations of different groups.

end

if $\text{sum of norm for group } i \geq \text{quantile}(\text{all groups}, \alpha)$ **then**

 | group i needs further splitting

else

 | group i does not need further splitting

end

Certain similar work has been done by others in the field[DDM⁺22]. They also used cosine similarity of data groups to evaluate whether we should further split a group. However, key differences exist between our proposed method and their idea:

They tend to split the groups based on hierarchical NMF until a certain fixed cosine similarity norm threshold is reached. We find layers of groups using Neural NMF and only further split on a group if the hard split using Neural NMF was not successful (i.e., results in zero entries in certain groups) and that the cosine similarity norm of one group is greater than a fixed quantile among the groups of a layer.

They perform cosine similarity on individual groups, i.e., evaluate whether a group needs further splitting by calculating $\cos(\mathbf{M}, \mathbf{M})$ given a matrix $\mathbf{M}$ and evaluating whether the Frobenius norm is greater than a fixed numeric threshold. Our method takes into account the pairwise relationship between various groups, and thus is more robust in finding latent relationships between groups.

Our method is likely to perform better because:

- It is difficult to determine what number we should use as the norm threshold, and even with a good choice, things differ for different layers of the tree structure in the data set. Choosing a quantile from 0 to 1 is much more convenient.
- Using the quantile method adjusts the threshold in a dynamic way, and we don't have to hard-cut anything.
- It is also worth noting that for Neural NMF it is nearly impossible to apply the idea used in Population Hierarchical NMF because it is a bottom-up method and splitting by a certain norm wipes out the different layers. HNMF, on the other hand, is a top-down method, which can be modified by using a certain norm threshold.

We also observed the difficulty for Neural NMF to group subgroups exactly as they should be, though rough groupings are guaranteed. We thus developed the method of clustering based on feature similarity. With the cosine similarity matrix defined before, we normalize the cosine similarity matrix by the size of the matrix $a \cdot b$. We define the cosine value as the Frobenius norm of the normalized cosine similarity matrix. Then we sum up the Frobenius norm of each group with respect to other groups. We then claim that if the maximum value (non-zero) of the list of cosine values for the group i is above a certain quantile threshold, we group i with that group. If two or more of the groups have a cosine value equal to zero with respect to any other group, they should be grouped together. Otherwise, the group should remain by itself.

A detailed pseudocode for this process is given below in the form of an algorithm.

Algorithm 2: Feature Similarity-Based Clustering

Input: A quantile level α .

A list L of m groups in the form of matrices of shape $c_m \times n$ (c the group size, n the fixed number of features) for cluster detection. At least one of the groups contains zero data points.

Result: A list of groups that needs further splitting.

for $i \in \{1, \dots, m\}$ **do**

for $j \in \{i + 1, \dots, m\}$ **do**

 Given i th matrix $\mathbf{A} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_a \end{bmatrix}$ and j th matrix $\mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_b \end{bmatrix}$, calculate $C_{pq} = \cos(a_p, b_q)$ for all

 combinations of p and q and obtain the cosine similarity matrix $\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1b} \\ \vdots & \vdots & \cdots & \vdots \\ C_{a1} & C_{a2} & \cdots & C_{ab} \end{bmatrix}$.

 Obtain normalized cosine similarity matrix $\begin{bmatrix} \frac{C_{11}}{a \cdot b} & \frac{C_{12}}{a \cdot b} & \cdots & \frac{C_{1b}}{a \cdot b} \\ \vdots & \vdots & \cdots & \vdots \\ \frac{C_{a1}}{a \cdot b} & \frac{C_{a2}}{a \cdot b} & \cdots & \frac{C_{ab}}{a \cdot b} \end{bmatrix}$.

 Calculate the Frobenius norm of the normalized cosine similarity matrix.

end

end

if norm for group i with group $j \geq \text{quantile}(\text{all norms for group } i, \alpha)$ **then**

 group i should be grouped with group j

else

 group i should group by itself

end

A combination of the two methods yields great improvements upon the results for Neural NMF, as demonstrated in Experiments.

4.4. NMF-Based Community Detection. In this section, we focus specifically on multiplex structure, looking into the potential of community detection by incorporating NMF and expanding upon the multiplex NMF-based community detection methods and their practical implications yielded through MyLy-meData applications. The NMF-based community detection methods on a multiplex graph of N layers delineated in this paper are an extension conflating community detection techniques with NMF with reference to the approach proposed in [GPZ16], in which the formulation of the problem involves NMTF (Equation 3.3) but in its symmetric form:

$$\mathbf{A}^{(i)} \approx \mathbf{H}\mathbf{S}^{(i)}\mathbf{H}^T, \quad \text{where } i \in \{1, \dots, N\} \quad (4.2)$$

Say we have n nodes and k communities. Then, $\mathbf{A}^{(i)} \in \mathbb{R}_+^{n \times n}$ is the symmetric adjacency matrix for layer i that is either binary or weighted, and $\mathbf{H} \in \mathbb{R}_+^{k \times n}$ is the community indicator matrix that is shared across all layers. Note that $\mathbf{H}, \mathbf{S}^{(i)}$ are nonnegative matrices. Furthermore, to overcome the unpredictability of unobserved entries, a binary weight matrix $\mathbf{W}$ is introduced to deal with missing values and is formed with respect to the following rules:

$$\forall i, \quad \mathbf{W}_{xy}^{(i)} = \begin{cases} 1 & \text{if } \mathbf{A}_{xy}^{(i)} \text{ is observed,} \\ 0 & \text{otherwise} \end{cases} \quad (4.3)$$

With a synthesis of previously discussed knowledge, it is explicated in [GPZ16] that $\mathbf{H}$ in Equation 4.2 can be determined through solving an optimization problem defined in 4.4 using a trade-off parameter

η_i to balance the equation:

$$\min \mathcal{L}(\mathbf{H}, \mathbf{S}^{(i)}) = \sum_{i=1}^N \|\mathbf{W}^{(i)} \circ (\mathbf{A}^{(i)} - \mathbf{H}\mathbf{S}^{(i)}\mathbf{H}^T)\|_F^2 + \sum_{i=1}^N \eta_i \|\mathbf{S}^{(i)}\|_1 \quad (4.4)$$

Approaches are provided in [GPZ16] for optimizing with natural gradient and finding expressions for multiplicative updates for $\mathbf{H}$ and $\mathbf{S}^{(i)}$, which are 4.5 and 4.6.

$$\mathbf{S}^{(i)} \leftarrow \mathbf{S}^{(i)} \circ \sqrt{\frac{\mathbf{H}^T (\mathbf{W}^{(i)} \circ \mathbf{A}^{(i)}) \mathbf{H}}{\mathbf{H}^T (\mathbf{W}^{(i)} \circ \mathbf{H}\mathbf{S}^{(i)}\mathbf{H}^T) \mathbf{H} + \frac{1}{2}\eta_i}} \quad (4.5)$$

$$\mathbf{H} \leftarrow \mathbf{H} \circ \left(\frac{\sum_{i=1}^N (\mathbf{W}^{(i)} \circ \mathbf{A}^{(i)}) \mathbf{H}\mathbf{S}^{(i)}}{\sum_{i=1}^N (\mathbf{W}^{(i)} \circ \mathbf{H}\mathbf{S}^{(i)}\mathbf{H}^T) \mathbf{H}\mathbf{S}^{(i)}} \right)^{\frac{1}{4}} \quad (4.6)$$

The community indicator matrix $\mathbf{H}$ is especially important, as it predicts how certain we should be to categorize each node as a member of a community. In this work, we extend the context of the aforementioned works by providing a full, effective procedure of determining $\mathbf{H}$ through multiplicative updates on an integration of layers. This procedure is demonstrated in Algorithm 3. The relevant helper functions can be readily imported from our repository. Within this community detection framework, we aim to ultimately construct a multiplex network of Lyme disease patients with adequate interpretability based on MyLymeData, which may inspire insights in performing exploratory analysis in such convoluted systems.

Algorithm 3: NMF-Based Multiplex Community Detection (developed upon [GPZ16])

Input: Dictionary $\mathcal{A}$ consisted of adjacency matrices $\mathbf{A}^{(i)} \in \mathbb{R}_+^{n \times n}$ for layer $i \in \{1, \dots, N\}$; Expected number of communities k ; Trade-off parameters η_i ; Tolerance value ϵ

Output: Community indicator matrix $\mathbf{H} \in \mathbb{R}_+^{n \times k}$

/* Procedure begins

*/

Randomly initialize $\mathbf{H} \in \mathbb{R}_+^{n \times k}$

Set empty dictionaries $\mathcal{S}$ and $\mathcal{W}$

for $i \in \{1, \dots, N\}$ **do**

 Randomly initialize $\mathcal{S}[i] = \mathbf{S}^{(i)} \in \mathbb{R}_+^{k \times k}$

 Set $\mathcal{W}[i] = \mathbf{W}_{n \times n}^{(i)}$, the binary weight matrix corresponding to $\mathbf{A}^{(i)}$ at layer i , using 4.3

end

while $E > \epsilon$ **do**

 Set $\mathbf{H}^* = \mathbf{H} \circ \left(\frac{\sum_{i=1}^N (\mathbf{W}^{(i)} \circ \mathbf{A}^{(i)}) \mathbf{H}\mathbf{S}^{(i)}}{\sum_{i=1}^N (\mathbf{W}^{(i)} \circ \mathbf{H}\mathbf{S}^{(i)}\mathbf{H}^T) \mathbf{H}\mathbf{S}^{(i)}} \right)^{\frac{1}{4}}$

 Set empty dictionary $\mathcal{S}^*$

for $i \in \{1, \dots, N\}$ **do**

 Set $\mathcal{S}^*[i] = \mathbf{S}^{(i)*} = \mathbf{S}^{(i)} \circ \sqrt{\frac{\mathbf{H}^T (\mathbf{W}^{(i)} \circ \mathbf{A}^{(i)}) \mathbf{H}}{\mathbf{H}^T (\mathbf{W}^{(i)} \circ \mathbf{H}\mathbf{S}^{(i)}\mathbf{H}^T) \mathbf{H} + \frac{1}{2}\eta_i}}$

end

 Set $E_H = \|\mathbf{H}^* - \mathbf{H}\|_F$

// Compute Frobenius norms for consecutive error

 Set $E_S = \frac{1}{N} \sum_{i=1}^N \|\mathbf{S}^{(i)*} - \mathbf{S}^{(i)}\|_F$

 Update $E = E_H + E_S$

 Update $\mathbf{H} = \mathbf{H}^*$ and $\mathcal{S} = \mathcal{S}^*$

end

/* Procedure ends

*/

4.5. Fairness-aware NMF for Dimension Reduction. Unlike commonly-used techniques for dimension reduction, like PCA, containing both positive and negative values in the decomposed matrices, NMF achieves dimension reduction by decomposing the original matrix into two non-negative matrices. After decomposition, the columns of one of the matrices can be regarded as the basis vectors. Then, we can obtain a dimension-reduced matrix by projecting the document vectors onto the lower dimensional space which is formed by these basis vectors.

Though PCA and NMF both effectively achieve the purpose of dimensionality reduction, it is found in real-world data that a certain level of unfairness is involved in this low-dimension representation, which can be introduced through the data itself and the algorithms as well. In specific, the problem of unfairness can be due to a skewed distribution of subgroup populations. For instance, in MyLymeData, after extracting the symptom dataset on unwell patients and preprocessing like described in 6, we use biological sex as the demographic attribute and obtain three subgroups: female, male, and 'other'. With a total population of 6901 patients: 5593 samples (81.05%) for 'female', 1298 samples (18.81%) for 'male', and only 10 samples (0.14%) for 'other'. The 'other' group is extremely underrepresented. Hence, when we perform dimension reduction on the data, the inequality of population sizes persists as shown in 22 below.

The group 'other' is suffering a much higher loss when compared to 'female' and 'male' groups. Hence, our goal for fair NMF is to mitigate this difference in population sizes and allow minority groups to be represented in a "fairer" fashion by sacrificing less loss in the low-dimension representation. In this report, we focus on developing fair NMF in terms of both rescaling data and embedding fairness constraints into the algorithm. We reformulate basic ideas of fairness from the paper [STM⁺18] that establishes a fair PCA algorithm.

Definition 4.1. (Reconstruction Error) NMF decomposes a non-negative data matrix $\mathbf{X}$ into two non-negative matrices $\mathbf{W}$ (basis matrix) and $\mathbf{H}$ (coefficient matrix), such that:

$$\mathbf{X} \approx \mathbf{WH}$$

We define the reconstruction error $\|\mathbf{X} - \mathbf{WH}\|_F$ as the difference between the original data matrix $\mathbf{X}$ and the reconstructed data matrix using NMF. The reconstruction error quantifies how well the factorized matrices $\mathbf{W}$ and $\mathbf{H}$ approximate the original data matrix $\mathbf{X}$. A lower reconstruction error indicates a better approximation, implying that the factorized matrices capture most of the underlying structure and patterns in the data.

In the context of NMF, reducing the reconstruction error is a primary goal. It ensures that the non-negative factors $\mathbf{W}$ and $\mathbf{H}$ accurately represent the original data while achieving dimensionality reduction. Researchers often use optimization techniques to minimize the reconstruction error, adjusting the values of $\mathbf{W}$ and $\mathbf{H}$ iteratively to improve the approximation.

Let $\mathbf{X} \in \mathbb{R}_+^{n_1 \times n_2}$ denote our data matrix where rows represent samples and columns represent features. The data consists of subgroups corresponding to different values of sensitive attributes. We will use two subpopulations A and B as the base case, and the results are generalizable to more than two groups. The optimization is formulated as:

$$\min_{\mathbf{W} \in \mathbb{R}_+^{n_1 \times r}, \mathbf{H} \in \mathbb{R}_+^{r \times n_2}} \max_{\mathbf{A}, \mathbf{B}} \left\{ \frac{\|\mathbf{A} - \mathbf{W}_A \mathbf{H}\|_F^2 - \|\mathbf{A} - \mathbf{A}^*\|_F^2}{|A|}, \frac{\|\mathbf{B} - \mathbf{W}_B \mathbf{H}\|_F^2 - \|\mathbf{B} - \mathbf{B}^*\|_F^2}{|B|} \right\} \quad (4.7)$$

where $|\cdot|$ denotes the sample size of a subgroup, $\mathbf{W}_A$ the restriction to the columns in $\mathbf{W}$ population A , and $\mathbf{A}^*$ is the optimal rank- r approximation of A computed by truncated SVD or non-negative rank- r approximation by NMF.

4.5.1. Fair NMF with Rescaling Data. We introduce a rescaling technique that assigns specific constants to each subgroup in order to effectively align all subgroups to reach the lowest value of loss possible.

Rescaling the data can lead to changes in the magnitude of the data points. This might affect the importance of certain features or components in the NMF decomposition. Suppose n subgroups from dataset $\mathbf{X}$ such that $X = [X_1, X_2, \dots, X_n]$. By applying this transformation, the new data becomes $[c_1 \cdot X_1, c_2 \cdot X_2, \dots, c_n \cdot X_n]$. The values of constants ensure that every subgroup receives fair consideration during dimension reduction.

Algorithm 4: Fair NMF with rescaling data (for two demographic groups)

Input: Data matrix $\mathbf{X}$ s where $\mathbf{X}s = \begin{bmatrix} \mathbf{A} \\ \mathbf{B} \end{bmatrix}$;

Output: Coefficient matrix $\mathbf{W}$ s; dictionary matrix $\mathbf{H}$; loss; optimal values for c

- implement vanilla NMF on $\mathbf{A}$ and $\mathbf{B}$ and compute their respective loss $\|\mathbf{A} - \mathbf{A}^*\|_F^2$ and $\|\mathbf{B} - \mathbf{B}^*\|_F^2$; compute loss for $\mathbf{A}$ and $\mathbf{B}$ as defined in **Definition 3.1**.
 - Rescale data using constant c_1 such that the new rescaled data will be $\begin{bmatrix} c \cdot \mathbf{A} \\ \mathbf{B} \end{bmatrix}$
 - Calculate the loss for each group in rescaled data with different values of c .
 - Search for the optimal value of c which results in the lowest equal values of loss for both groups $\mathbf{A}$ and $\mathbf{B}$.
-

Our objective is to minimize the maximum loss suffered by any group, so by manipulating the value of c , it will change the weight on the highest loss suffered by any group. Consequently, the optimization problem could be different from the original one.

4.5.2. *Preliminary Fair NMF.* The preliminary fair NMF algorithm 5 is motivated by the min-max nature of the optimization problem and the fact that the factorization and the loss defined for each group can be computed separately. The high-level idea is that after the initialization, we compute the loss of each group and find the group with the highest loss. Then we apply one iteration of the multiplicative update to that group and repeat the process.

Algorithm 5: Preliminary Fair NMF (for two demographic groups)

Input: Data matrix $\mathbf{X}$ s where $\mathbf{X}s = \begin{bmatrix} \mathbf{A} \\ \mathbf{B} \end{bmatrix}$; number of features; learning rate η ; maximum number of iterations

Output: Coefficient matrix $\mathbf{W}$ s; dictionary matrix $\mathbf{H}$; loss; minimized reconstruction error for subgroups

- Randomly initialize the dictionary matrix $\mathbf{H}$ for all groups and the coefficient matrix $\mathbf{W}_\mathbf{A}$ and $\mathbf{W}_\mathbf{B}$ for each group.
 - Compute their respective minimized reconstruction errors $\|\mathbf{A} - \mathbf{A}^*\|_F^2$ and $\|\mathbf{B} - \mathbf{B}^*\|_F^2$; compute loss for $\mathbf{A}$ and $\mathbf{B}$ as defined in **Definition 3.1**.
 - Find the group with the highest loss and denote it as group $\mathbf{G}$.
 - Update $\mathbf{W}$ s and $\mathbf{H}$ using the multiplicative update rule in [LS00] for the subgroup i with the highest reconstruction loss.
 - (1) $\mathbf{W}_\mathbf{G} \leftarrow \mathbf{G} \odot \left(\frac{\mathbf{G}\mathbf{H}^T}{\mathbf{W}_\mathbf{G}\mathbf{H}\mathbf{H}^T} \right)^\eta$
 - (2) $\mathbf{H} \leftarrow \mathbf{H} \odot \left(\frac{\mathbf{W}_\mathbf{G}^T \mathbf{G}}{\mathbf{W}_\mathbf{G}^T \mathbf{H}} \right)^\eta$
-

4.5.3. *Fair NMF with Weighted-Average Dictionary Matrix.* After implementing the preliminary version of fair NMF on synthetic data, we observe the fluctuations of subgroup losses when we switch from one group to another in the update process.

It suggests that the previous update rule is changing the dictionary matrix $\mathbf{H}$ too much from optimizing one group to another, as $\mathbf{H}$ is shared among all subgroups. To rationalize this, when one subgroup is

being updated, the $\mathbf{H}$ matrix is subject to changes for the entire population. We think that in the mean-time of updating $\mathbf{H}$ for one group, the other subgroups suffer unnecessary changes, which causes the losses for those groups to fluctuate.

To further address the instability of the dictionary matrix $\mathbf{H}$ between and within groups, we decide to explore the idea of using a reweighted $\mathbf{H}$ matrix by taking the previous $\mathbf{H}$ into account. Our approach is to capture the n numbers of the most recent $\mathbf{H}$ and reweight them. In our case, we are using the three most recent $\mathbf{H}$ with an emphasized weight on the latest one. The new matrix will then replace the oldest one among the three.

By keeping the history of the previous dictionary matrices, we hope it can help smooth out drastic changes whenever a new group is fed into the optimization algorithm. Hence, we propose an improved version of the preliminary fair NMF algorithm - Fair NMF with a Weighted-Average Dictionary Matrix. We are showing two subgroups below, but our algorithms can be generalized to have more groups.

Algorithm 6: Fair NMF with Weighted-Average Dictionary Matrix (for two demographic groups)

Input: Data matrix $\mathbf{X}$ s where $\mathbf{X}s = \begin{bmatrix} \mathbf{A} \\ \mathbf{B} \end{bmatrix}$; number of features; learning rate eta η ; maximum number of iterations

Output: Coefficient matrix $\mathbf{W}$ s; dictionary matrix $\mathbf{H}$; loss; minimized reconstruction error for subgroups

- implement vanilla NMF on $\mathbf{A}$ and $\mathbf{B}$ and compute their respective minimized reconstruction errors $\|\mathbf{A} - \mathbf{A}^*\|_F^2$ and $\|\mathbf{B} - \mathbf{B}^*\|_F^2$; compute loss for $\mathbf{A}$ and $\mathbf{B}$ as defined in **Definition 3.1**.
 - Find the group with the highest loss and denote it as group $\mathbf{G}$.
 - update $\mathbf{W}$ s and $\mathbf{H}$ using the multiplicative update rule in [LS00] for the subgroup i with the highest reconstruction loss.
 - (1) $\mathbf{W}_G \leftarrow \mathbf{G} \odot \left(\frac{\mathbf{G}\mathbf{H}^T}{\mathbf{W}_G\mathbf{H}\mathbf{H}^T} \right)^\eta$
 - (2) $\mathbf{H} \leftarrow \mathbf{H} \odot \left(\frac{\mathbf{W}_G^T\mathbf{G}}{\mathbf{W}_G^T\mathbf{G}\mathbf{H}} \right)^\eta$
 - keep track of the three most recent $\mathbf{H}$ and reweight them with the highest weight given to the most recent one to generate $\hat{\mathbf{H}}$; remove the oldest $\mathbf{H}$ among the three and append $\hat{\mathbf{H}}$; iterate the process to update $\hat{\mathbf{H}}$.
-

Hyperparameter optimization is also an important aspect of our fair NMF algorithm. We look closer at η , which is the step size in the multiplicative update of NMF. It controls the magnitude of the updates being applied to the matrices $\mathbf{W}$ s and $\mathbf{H}$ during our iterative optimization process. We can also perceive η as the learning rate for getting a fair representation among subgroups. Our adjustment to η involves finding a balance between subgroup convergence and descent of reconstruction loss. One attempt is to schedule our learning rate so that it optimizes for both purposes. When η is large, it encourages faster convergence but leads to instability. On the other hand, when η is small, even though the convergence slows down, it reduces the risk of skipping solutions and stabilizes the descent of our reconstruction errors. In specific, for the first half of our maximum iterations, we implement a much faster learning rate to promote convergence by having a η multiplier. Then, we have a decaying learning rate schedule for the second half of the iterations to have a more stable descent of subgroup losses.

4.5.4. Early Stopping Strategy. In Figure 2, it is shown that in running the Fair NMF algorithm, there are certain iterations after which the difference between the losses of each group is small. However, in later iterations, as the group is updated for shifts, the difference in the losses increases again. Therefore, we propose an early stopping algorithm in order to stop the algorithm as soon as a fair NMF decomposition is obtained.

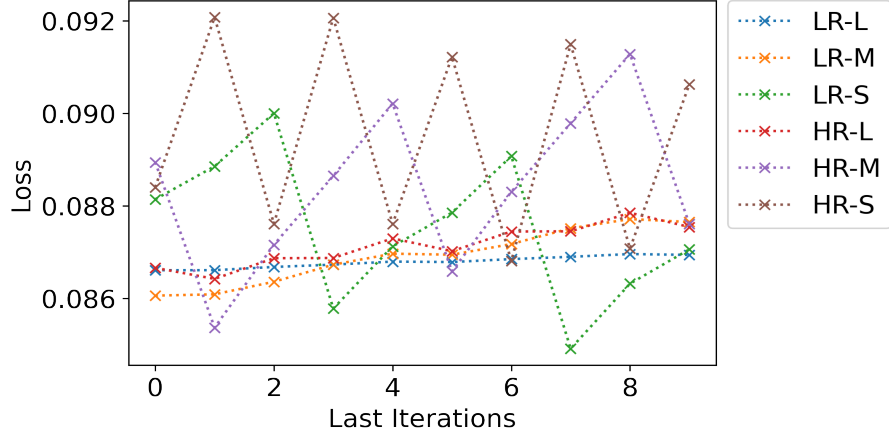


Figure 2 Fair NMF loss: this plot is obtained by running the preliminary fair NMF algorithm 5. The total iterations ran is 6000, and the last ten iterations are plotted. The y-axis is the loss as defined in equation 4.7. The losses for the six groups of the synthetic data as defined in section 5.3 are plotted.

In algorithm 7, the group with the highest loss is updated until the loss of another group surpasses the current group. Then for every k iteration, the pairwise difference between the losses of each group is checked. If each pairwise difference is smaller than a specified tolerance ϵ , the algorithm stops. To obtain the pairwise difference, we compute the difference of losses between one group and all the rest groups and then compare all the losses to the tolerance. If there are n groups, $\frac{n(n-1)}{2}$ differences are computed. As n has higher values, computing pairwise differences will be time-consuming. Alternatively, we can compute the loss of each group and find the difference between the maximum and minimum of all. If the difference between the maximum and minimum is smaller than ϵ , the algorithm stops.

Algorithm 7: Early-Stopping Fair NMF

Input: Data matrix $\mathbf{X}$ s where $\mathbf{X}s = \begin{bmatrix} \mathbf{A} \\ \mathbf{B} \end{bmatrix}$; number of features; learning rate η ; maximum number of iterations; k number of iterations between each early-stopping check; ϵ the tolerance for the difference of loss between groups; maximum number of iterations N .

Output: Coefficient matrix $\mathbf{W}$ s; dictionary matrix $\mathbf{H}$; loss; minimized reconstruction error for subgroups

```
/* Procedure begins */
Run the vanilla NMF and compute the loss for each group.
 $G \leftarrow$  is the group with the greatest loss.
 $m \leftarrow 1$ 
while  $m < N$  do
    Update  $H$  with respective to  $G$  using multiplicative update.
    if  $m \% k = 0$  then
        Use non-negative least squares to recover the coefficient matrix for the groups that have not been updated.
        Get pairwise loss difference between groups.
    end
    if each pairwise difference  $< \epsilon$  then
        break
    end
    else
        Update  $G$  as the group now with the highest loss
    end
     $m \leftarrow m + 1$ 
end
/* Procedure ends */
```

5. SYNTHETIC EXPERIMENTS

5.1. Neural NMF. A visual demonstration of the synthetic data set used to run tests is demonstrated as follows. Starting from one main topic, each of the topics branches out into two evenly split topics of the same size. In the following layers, the subtopics split further, creating more branches and smaller topics. It is worth noting that the last layer of groups is split randomly into two from the previous layer.

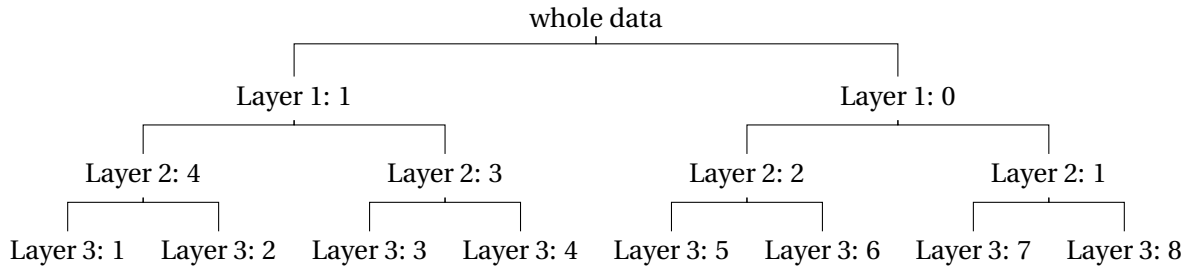


Figure 3 Demonstration of the synthetic data in a tree diagram. The value of a data point at a specific level is encoded as the number after the Layer number. For instance, "Layer 3: 2" indicates that the data point has a value of 2 at the third layer.

To be more specific, we structured the data in such a way that out of 200 data points, half at layer one has value 1 and the other half has value 0. Out of the 100 data points in layer 1 with value 1, half of the

data points in the second layer have value 4, and half have value 3. Similarly, half of the data points with a layer 1 value of 0 have a second layer value 2 and the other half has value 1. For the third layer, we have each of the layer 2 groups, each group containing data points with the same value in layer 2, divided into groups of 2 distinct values in layer 3. We then processed the data by hot-encoding the data points.

The table below gives the results of 50 runs on this synthetic data set for the feature similarity-based cluster detection method in comparison to the fixed quantile threshold method.

type of result (total of 5 types)	total count for our method	total count for fixed thresholds
Correct	13	3
Partially Correct	29	10
Partially Correct and Partially incorrect	3	27
No Further Split	1	9
Incorrect	4	1

For further clarification, the term "Correct" means that the algorithm identified every group that needs further splitting with exact precision. The term "Partially Correct" indicates that the algorithm identified some of the groups that need further splitting, with no wrong attempts. "Partially Correct and Partially incorrect" means that some right and wrong potential groups that need further splitting were detected. Accuracy does not change much in this case. "No Further Split" indicates that the algorithm made no apparent change to the current situation, i.e., made no attempt to further split. Last but not least, "Incorrect" means that everything is wrongly identified and this results in a definite loss in accuracy.

We also tested our results for the feature similarity-based clustering method. Given the exact groups of the bottom most layer of the data set, we ran 100 runs on the random bottom most groups and resulted in 100% accuracy. With unevenly tree-structured data with lopsided branching, the algorithm resulted in a 96% accuracy.

Combining the two methods together, we test the accuracy of the grouping of the second layer in our tree-structured data. The accuracy of unsupervised vanilla Neural NMF on synthetic data for the second layer is 58.13%. Using preliminary results for the bottom most layer of groups for Neural NMF, we first perform the further splitting method to get more accurate groups. Then we perform the grouping method on the modified groups to get the second layer of groups, which yields an accuracy of 74.56%.

5.2. NMF-based Community Detection. On a synthetic 3-layer network with 3 ground truth communities as shown in Figure 4, the NMF-based community detection method yields perfect clustering accuracy scores on multiplex structure as shown in Table 1. The probabilities of community assignment for all nodes, derived from the normalized resulting community indicator matrix $\mathbf{H}$, is visualized in Figure 5e.

	Homogeneity	Normalized Mutual Info (NMI)	Random Index (RI)
Multiplex Structure	1.0	1.0	1.0
Monoplex Structure	0.64756	0.64703	0.84558

Table 1 Clustering accuracy scores of organizing the synthetic network in monoplex structure (Figure 5d) and multiplex structure (Figure 5e).

Through applying Algorithm 3 to different compositions of data, we derive Figure 5 and Table 1 which demonstrate the advantage of modeling the data as a multi-layer network. Compared with the super-imposed monoplex structure, the multiplex structure leads to higher clustering accuracy and is more

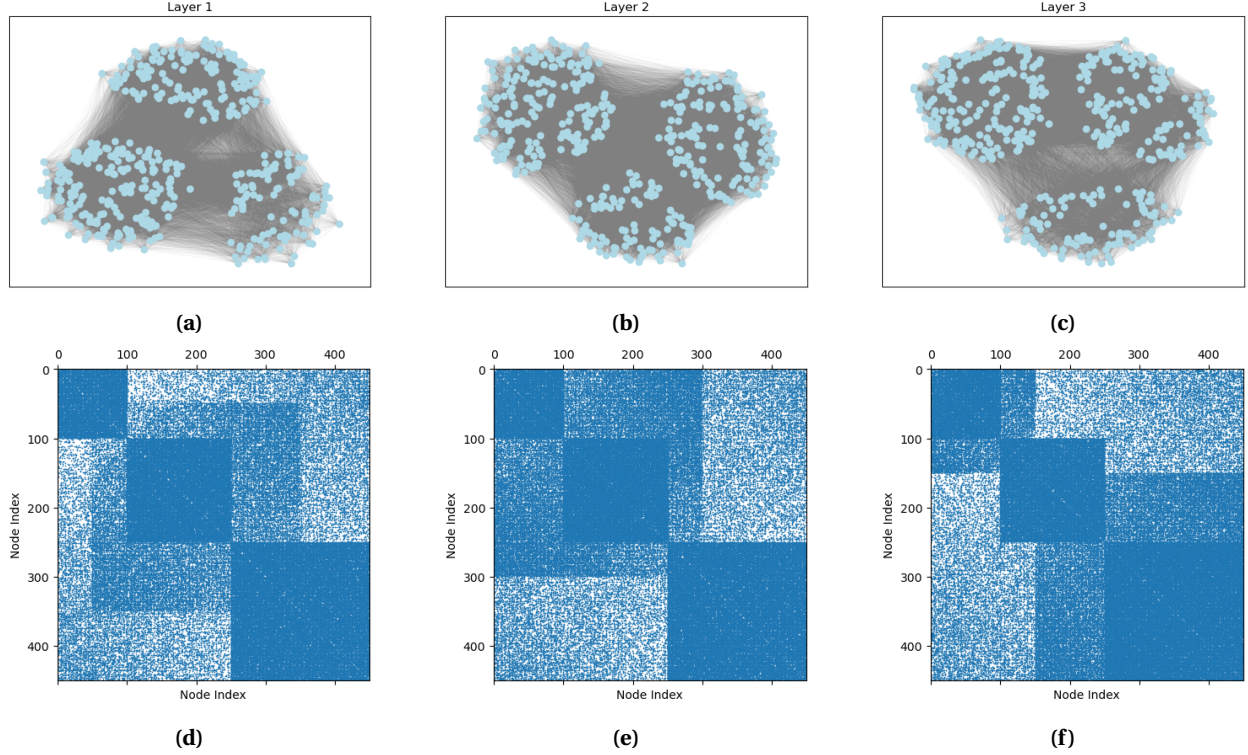


Figure 4 The network graphs and spy plots of the synthetic multiplex network with 450 nodes, 3 layers, and 3 ground truth communities containing 100, 150, and 200 nodes respectively are shown above. Specifically, Figure 4a and 4d visualize layer 1, Figure 4b and 4e visualize layer 2, and Figure 4c and 4f visualize layer 3. The layers are accompanied by different patterns of noise but with similar density levels that are used to test the robustness of the algorithm. The network graphs are generated using NetworkX.

insusceptible to noise. It produces a more unified probability distribution for nodes belonging to a common ground truth community. In general, the NMF-based multiplex community detection algorithm performs well with a clustering accuracy ideal in synthetic experiment like this and acceptable in most other cases.

5.3. Fair NMF. We created a synthetic dataset that looks at variations in nonnegative rank and population sizes. Several parameters are defined to control the characteristics of the synthetic data:

- m_1, m_2 , and m_3 : These variables represent the number of samples for three different sizes respectively small, medium, and large.
- n : The number of features or dimensions for each dataset.
- d_1, d_2 : These variables represent the intrinsic or underlying dimensions of the low-rank (d_1) and high-rank (d_2) components of the synthetic data.

Then we generate the synthetic data. $\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3, \mathbf{X}_4, \mathbf{X}_5$, and $\mathbf{X}_6$ are numpy arrays that represent different synthetic datasets with numbers between 0 and 1.

- $\mathbf{X}_1$ is a large-sized dataset with low rank.
- $\mathbf{X}_2$ is a medium-sized dataset with low rank.
- $\mathbf{X}_3$ is a small-sized dataset with high rank.
- $\mathbf{X}_4$ is a large-sized dataset with high rank.
- $\mathbf{X}_5$ is a medium-sized dataset with high rank.

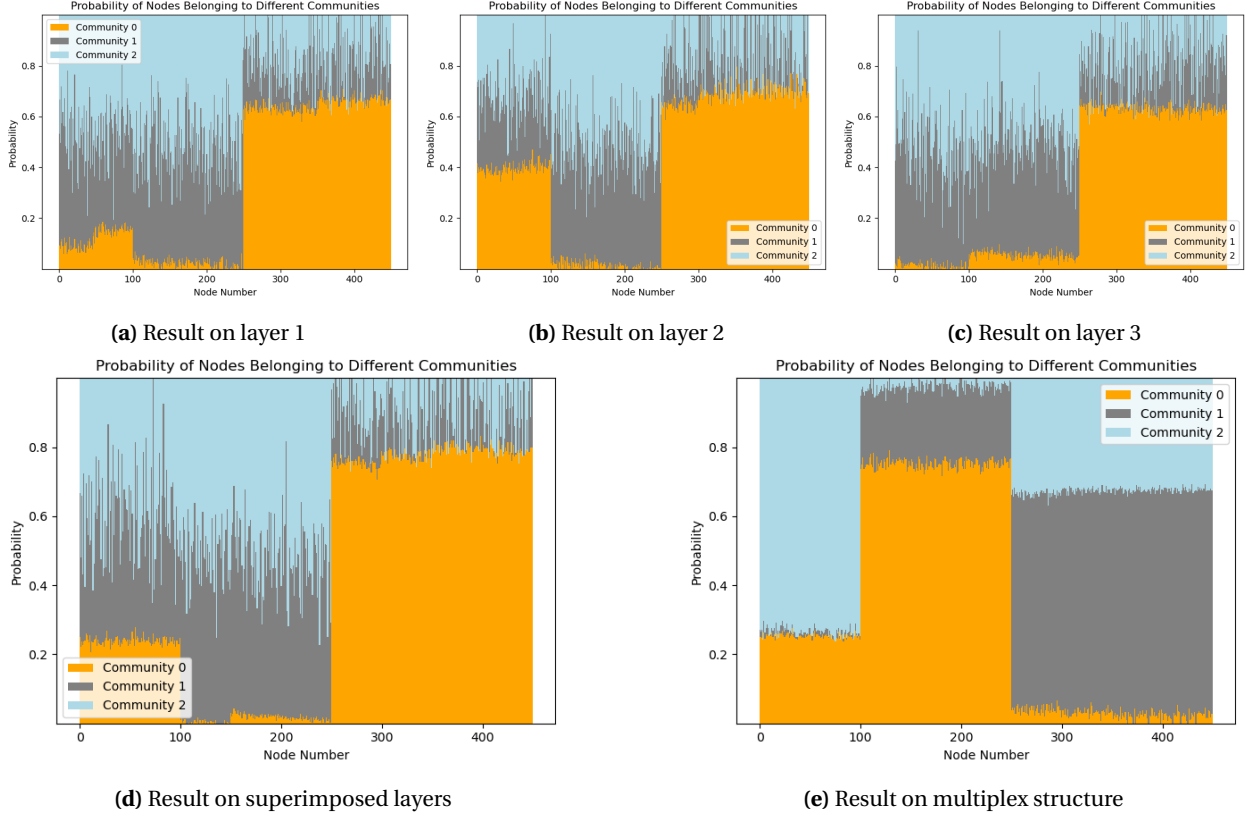


Figure 5 The 5 stacked bar plots are generated using their corresponding output community indicator matrix $\mathbf{H}$. Entries in $\mathbf{H}$ are normalized with respect to row. The more proportion a color takes in a stacked bar, the more likely this node belongs to the corresponding community. Specifically, 5a, 5b, and 5c show the results of community detection performed on each single layer. 5d shows the community detection results on monoplex structure formed by superimposition of the 3 layers. 5e shows the result of implementing on multiplex structure, where the nodes are more stably divided into 3 distinctive groups consisting of nodes labeled 0-99, 100-249, and 250-449 respectively.

- $\mathbf{X}_6$ is a small-sized dataset with low rank.

Later in all the plot legends, we use the following abbreviation names to refer the six groups:

- Large size low rank: LR-L
- Medium size low rank: LR-M
- Small size low rank: LR-S
- Large size high rank: HR-L
- Medium size high rank: HR-M
- Small size high rank: HR-S

We specify these values for each experiment. Furthermore, whenever we refer to the rank of matrix, we're referring to nonnegative rank.

5.3.1. Rescaling data. We decided to rescale data in order to observe which rescaling constant would result in a fairer dimensionality-reduced representation of the point. The loss for group **A** and **B** diverges for values of c_1 smaller than 0 and higher than 3, so we search for $c \in [0, 3]$. Figure 6 shows how the value of loss changes with different values of c_1 . $c_1 = 1$ means vanilla NMF is executed, and for different values of c_1 , we're changing the value of the maximum loss suffered by any group before solving the loss

optimization problem. In figure 7, we plotted rescaled groups A and B for various ranks 3, 6, 9, and 12, and then searched for the best value for c_1 .

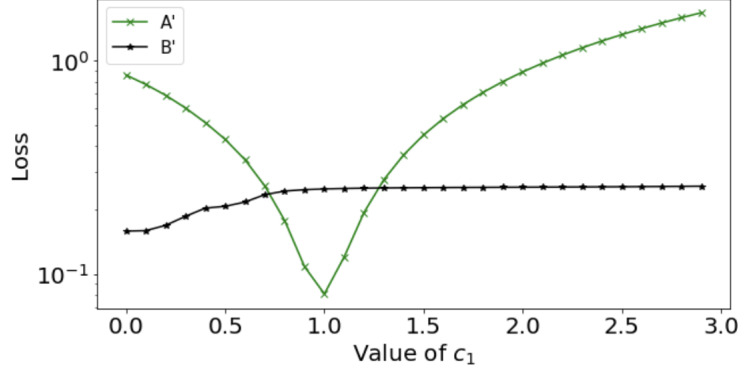


Figure 6 Using Scaled data with two subgroups. Number of components for NMF = 3. A' denotes rescaled group A , and B' denotes rescaled group B . Description of synthetic data: $|A| = 30$, $|B| = 10$, number of features = 20, $A_{rank} = 4$, $B_{rank} = 10$.

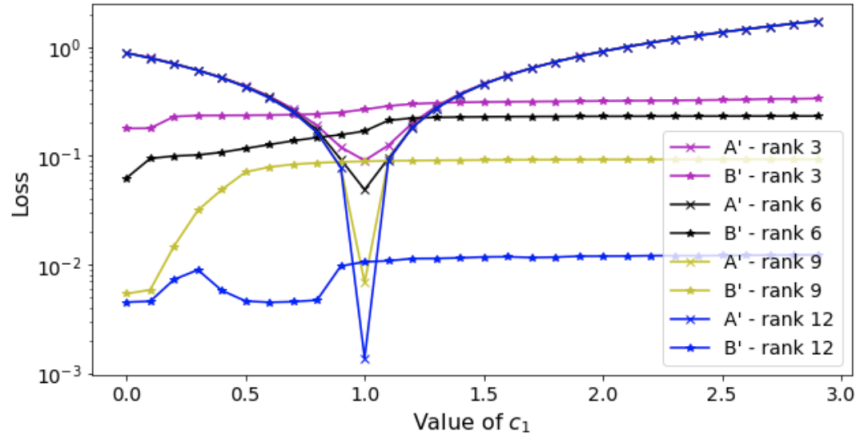


Figure 7 Using Scaled data with two subgroups with ranks 3, 6, 9, and 12. A' denotes rescaled group A , and B' denotes rescaled group B . Description of synthetic data: $|A| = 30$, $|B| = 10$, number of features = 20, $A_{rank} = 4$, $B_{rank} = 10$.

Our goal in rescaled data approach is to search for the value of c which result in the lowest and closest loss suffered by both groups. One of the advantages of our approach lies in its scalability, making it applicable to datasets of varying sizes and complexities. Moreover, by embracing fairness as a guiding principle, our method could be ideally suited for real-world applications where data from diverse populations or demographics need to be analyzed without bias.

5.3.2. Preliminary fair NMF. We first did experiments on the preliminary fair NMF shown in algorithm 5 and obtained the results shown in figure 10 and 11. To further inspect the preliminary results, we produced figure 8 that show the trade-off between each group at the end of iterations. To address the alternating trade-off problem, we developed the fair NMF with weighted-average dictionary matrix 6 and 7 and ran experiments on these methods. The results are shown in figure 12, 9. We mainly inspected on how the losses vary according to the rank used in the factorization and how the losses of different groups are changing given a fixed rank.

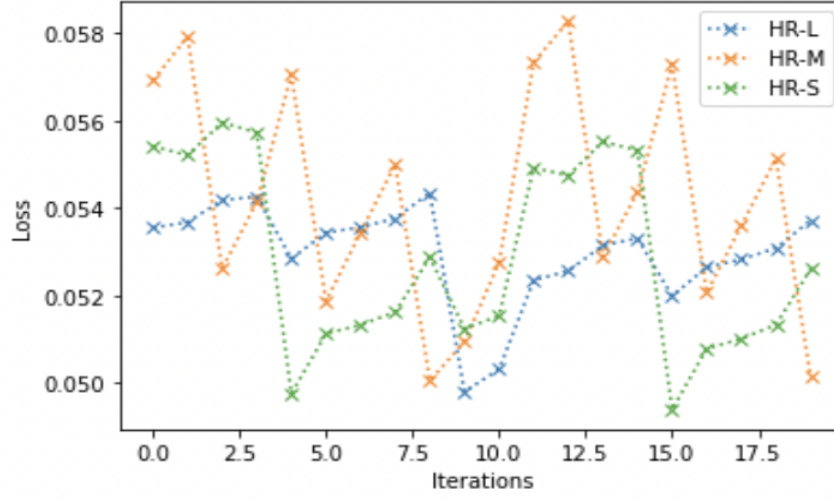


Figure 8 Fluctuations of Losses After Preliminary Fair NMF: this plot shows the losses of the three high-rank groups in the last 20 iterations out of 3000 iterations in total running algorithm 5. The rank used for factorization is 20.

5.3.3. *Fair NMF with Weighted-Average Dictionary.* The synthetic dataset to generate figure 9, 10, 11 has $m_1 = 1000$, $m_2 = 500$, $m_3 = 250$, $d_1 = 3$, $d_2 = 6$, and $n = 20$. We run an experiment on algorithm 5 with varying ranks for factorization to see the behavior of the losses defined in formulation 4.7 and the reconstruction error of NMF.

First notice that since the $\mathbf{A}^*$ is obtained through truncated SVD, and the highest rank is 6, $\|\mathbf{A} - \mathbf{A}^*\|$ will be 0 when the rank used in the NMF is greater than 6. Thus, figure 10 and 11 perform the same way for ranks greater than 6.

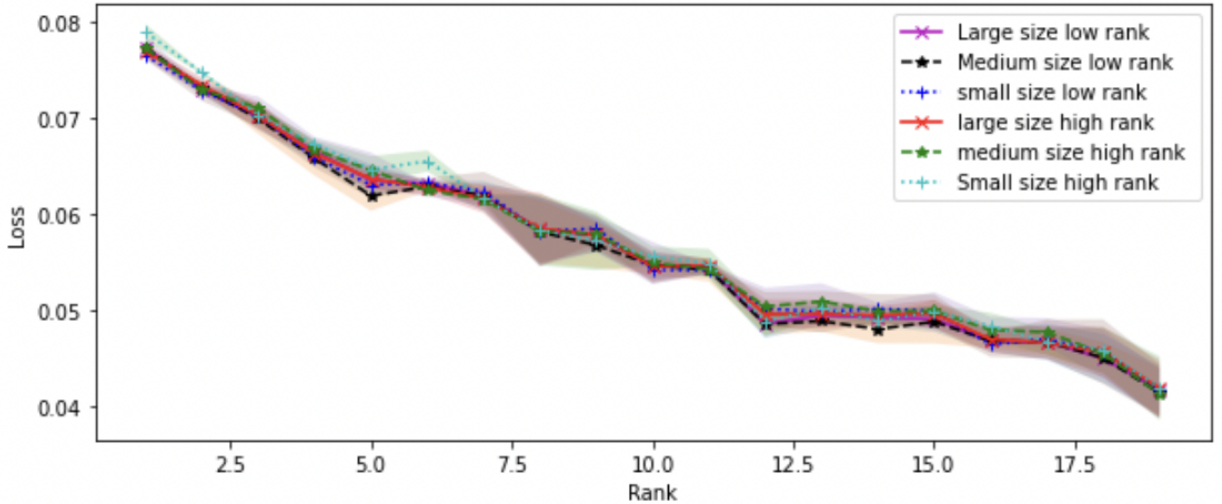


Figure 9 Loss of Fair NMF with Reweighted Dictionary Matrix (log scale): algorithm 6 is run for rank from 1 to 20. The loss is plotted here after obtaining the final factorization is plotted for each rank. The method to approximate the optimal factorization is truncated SVD.

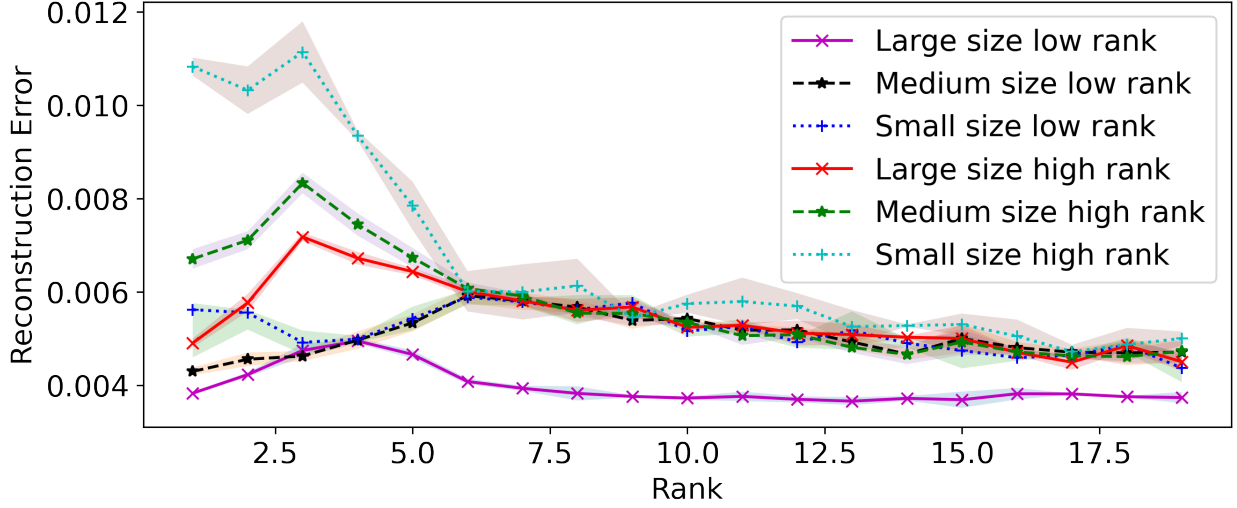


Figure 10 Reconstruction Error of Fair NMF (Truncated SVD, Squared Scale): algorithm 5 is run for rank from 1 to 20. The reconstruction error plotted here is defined as $\|\mathbf{X} - \mathbf{WH}\|_F^2$ after obtaining the final factorization is plotted for each rank. The method to approximate the optimal factorization is truncated SVD.

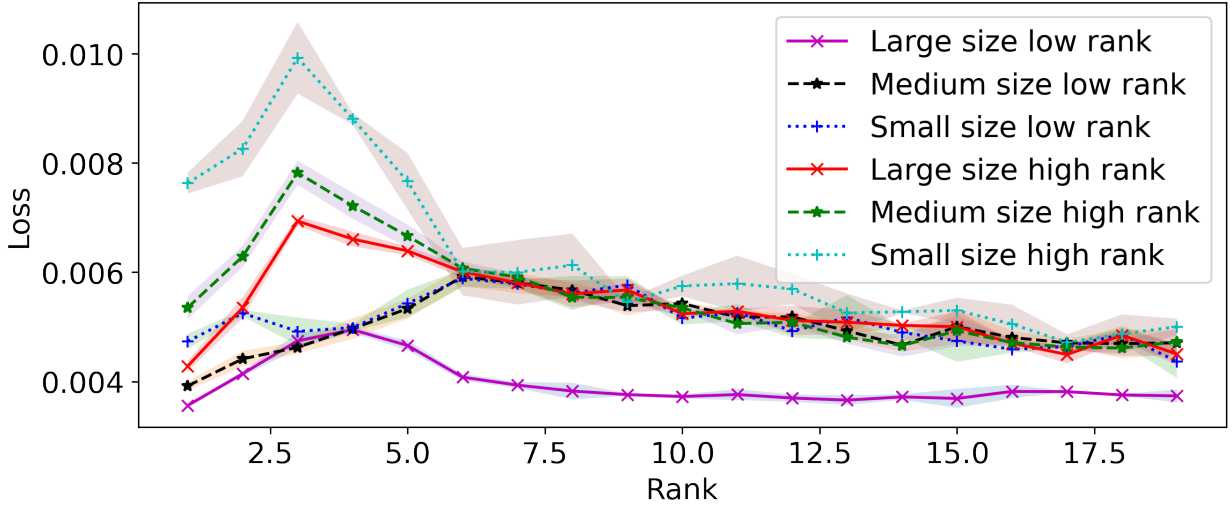


Figure 11 Loss of Fair NMF (Truncated SVD, Squared Scale): algorithm 5 is run for rank from 1 to 20. The squared loss defined in 4.7 is plotted. The method to approximate the optimal factorization is truncated SVD.

Note that the large size low-rank group always outperforms all the rest groups. When the rank grows from 4 to 5, the loss of the groups of small and medium size with low rank increases, while the groups of high ranks decrease. However, notice that the loss of large size group of low rank is still decreasing despite that the loss for other low-rank groups is increasing.

To explore the cause of why the large size low-rank group outperforms all the other groups and why the losses do not decrease when the rank is large, we run the experiment on a fixed rank of 20 and plot the losses of each group in the last 20 iterations. The result is shown in figure 8, and we can see the trade-off of losses between the groups is drastic. For example, at the 15th iteration plotted, the losses for both high-rank large size and high-rank small size groups decrease, but the loss of the high-rank medium size

group spikes. In hoping to smooth out the trade-offs, we develop the fair NMF with weighted -average dictionary method.

5.3.4. Early-Stopping. The synthetic dataset used for testing and experimentation in early-stopping fair NMF the sample sizes $m_1 = 1000$, $m_2 = 500$, and $m_3 = 250$. The high-rank group has a non-negative rank $d_1 = 6$, and the low-rank group has a non-negative rank $d_2 = 3$. And the synthesized data has $n = 40$ features.

We run the experiment of early stopping NMF on synthetic data. There are six groups in the synthesized matrix X as shown in the legends of figure 12 and 13. The rank that the algorithm is run on is 30 and the tolerance for the difference between losses is 0.05.

From figure 12 and 14, we can see that the number of iterations is successfully decreased by algorithm 7. Given that both experiments are run given $k = 3$, only 48 iterations are run for figure 12, and 27 for 14.

However, the trade-off of losses between groups does not happen in the expected way. From figure 12 and 13, we can see that the trade-off of losses mainly happens in the groups of small and medium sizes. Meanwhile, especially, the group with low rank and large size almost does not sacrifice even though it is not frequently updated.

To check the early-stopping algorithm's performance on finding an NMF with a lower rank, the algorithm is run for rank three, and the results are shown in figure 14 and 15. When running on a smaller rank, the similar behavior of the losses as described for figure 12 and 13 is still present. Notice that it is confusing that the losses of different groups are different, at initialization even though the initialization is completely random.

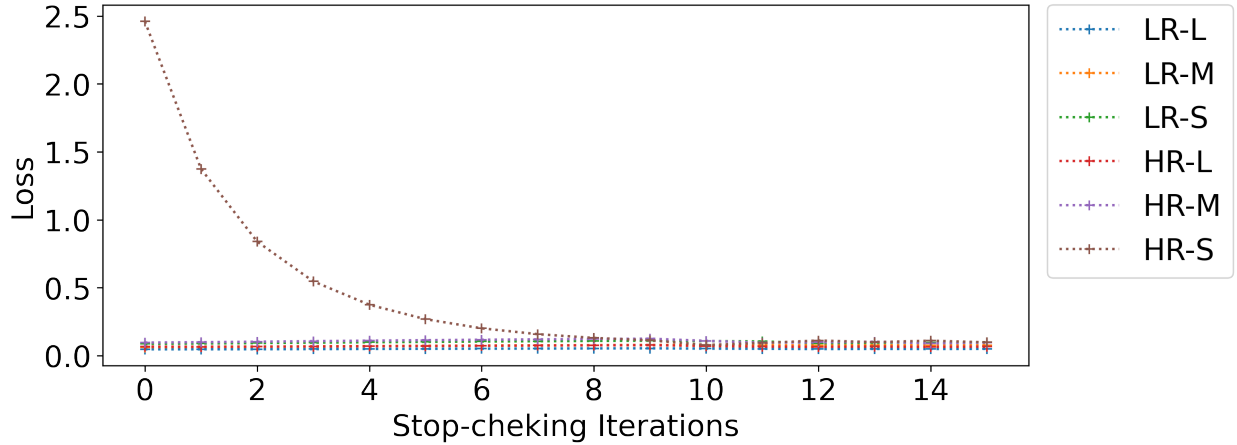


Figure 12 Early-Stopping Fair NMF Loss: this plot is generated by running algorithm 7 with $k = 3$ and $\epsilon = 0.05$. The rank that is used to decompose the input matrix is 30. The synthetic data used is as specified in 5.3 The x-axis shows the iterations in which the current difference between losses of each group is computed and compared to the tolerance. The y-axis shows the loss as defined in 4.7.

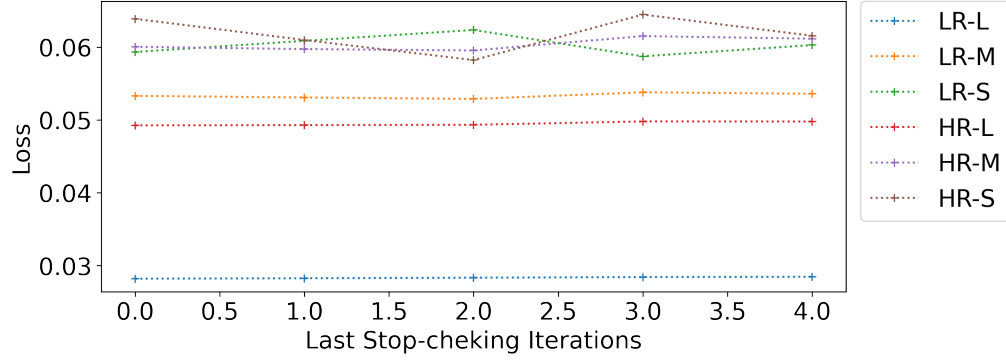


Figure 13 Early-Stopping Fair NMF Loss for the last 20 iterations: this is a zoom-in of Figure 12, in which the last five stop-checking iterations are plotted. The synthetic data used and the x,y-axes are the same as figure 12.

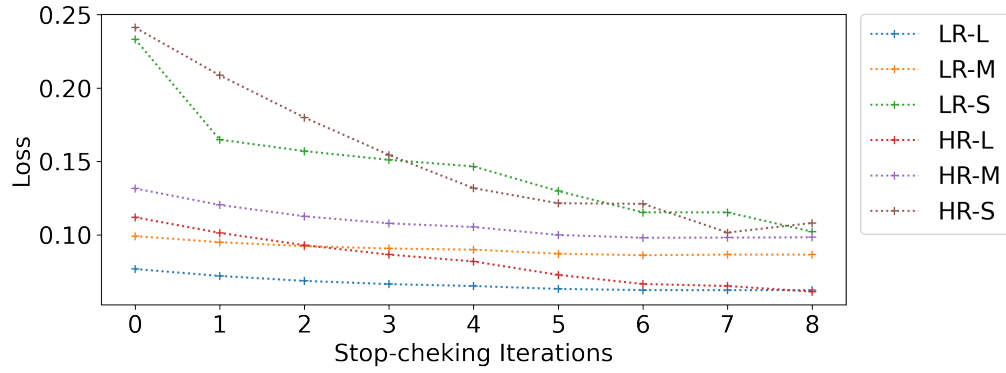


Figure 14 Early-Stopping Fair NMF Loss (Rank 3): this plot is generated by running algorithm 7 with $k = 3$ and $\epsilon = 0.05$. The rank that is used to decompose the input matrix is 3. The synthetic data used and the x,y-axes are the same as figure 12.

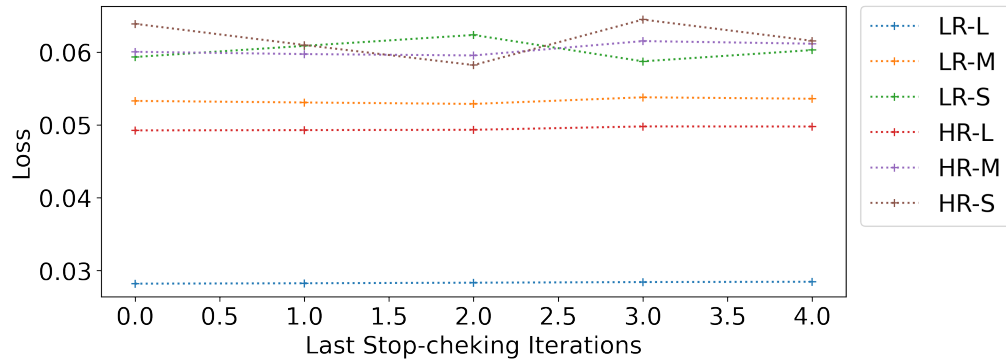


Figure 15 Early-Stopping Fair NMF Loss Last Iterations (Rank 3): this is a zoom-in of figure 12, in which the last five stop-checking iterations are plotted. The synthetic data used and the x,y-axes are the same as figure 12.

6. LYME DATA PREPROCESSING

We first removed 'other' values from: questions about income, time after the onset of symptoms before an accurate diagnosis, and current stage of one's Lyme disease.

Following this, we resolved an error in the initial Lyme coding. For questions about the severity of various Lyme symptoms (Current Severity of Symptoms and Severity of Symptoms when Lyme disease was at its worst), the severity ranged from 1-7 (1. No problem 2. Minimal problem 3. Mild problem 4. Moderate problem 5. Moderately severe problem 6. Severe problem 7. Very severe problem). We recorded the number associated with each rating, by replacing values of 7 with 5, values of 5 with 6, and values of 6 with 7, so the numerical data was ordinal. The severity range in our code now reads (1. No problem 2. Minimal problem 3. Mild problem 4. Moderate problem 5. Moderately severe problem 6. Severe problem 7. Very severe problem).

Next, we for the question about co-infections one was diagnosed with when diagnosed with Lyme, we combined 1 (diagnosed with supporting labs) and 2 (diagnosed without supporting labs). We assume that the distinction between co-infections with or without supporting labs is due to a difference in money the patient had to perform lab tests, rather than a significant difference in the co-infection itself.

For the current status question, asking a participant their current status in regards to Lyme disease, we coded "Sick or ill", "Neither sick nor well and take treatment" and "Neither sick nor well but not on treatment" as unwell. Additionally, we coded those who answered they were well or in remission as well. Combining various answers into a general 'unwell' label allowed us to have a binary representation of well/unwell patients.

Finally, for a question about the current stage of the participants' Lyme disease, we removed participants who selected "Tick bite occurred but no symptoms", "Don't know", and "Other". We coded "Initial Erythema Migrans rash" and "early or disseminated Lyme disease" as Early Lyme disease. For this question, the relevant comparisons were between participants with Early Lyme disease, chronic Lyme disease, and late-stage untreated Lyme disease.

6.1. Preprocessing for NMF on Symptoms. To work with NMF and study underlying patterns in symptoms between well and unwell patients, we removed patients who skipped eight or more questions as there are thirteen questions in total including an option for choosing other symptoms. And we replaced missing data with 1 which stands for no problem.

6.2. Preprocessing for SSNMF on Neurological Manifestations. To prepare the data for computing SS-NMF and study indicators of neurological manifestations, we include the time of diagnosis, prior misdiagnosis, and tick-borne co-infections including five different types of co-infections in our analysis. Questions related to tick bites are factors we have tested on, for example, whether a patient recalls a tick bite, the length of time the tick had been attached, whether the patient was treated with antibiotics and the length of time the patient was treated for the tick bite. We studied the group who identified themselves as unwell patients with chronic Lyme disease. For preprocessing and model computational purposes, we removed patients we answered "I don't know" or "prefer not to answer" in our analysis and replaced missing data with 0. We also normalized our data to a standard scaling from 0 to 1 using MinMaxScaler from scikit-learn [PVG⁺11] package.

7. MYLYMEDATA EXPERIMENTS

7.1. NMF Implementation on Symptoms Questions. In this section, we explore how symptoms contribute to the status of a patient, well or unwell, by implementing standard NMF on symptoms. Following the preprocessing steps in section 6.1, we used unsupervised NMF separately on well and unwell patients. In Figure 16, we computed using the **A** matrix, the dictionary matrix which represents how important each feature is for each topic, for both groups regarding symptoms questions. We have 12 symptoms in total and a single option for other symptoms excluded in the NMF computation. We have

chosen 5 topics and studied the answers from well and unwell patients with rating scales from 1 to 7 based on severity, where 1 stands for no problem and 7 stands for very severe problems.

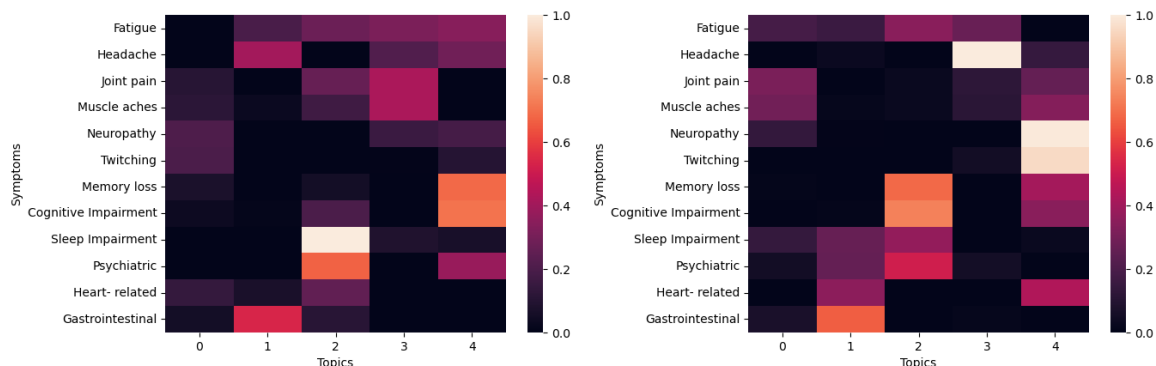


Figure 16 Left: heatmap of A matrix (feature by topics) for well patients based on the severity of symptoms. Right: heatmap of A (feature by topics) matrix for unwell patients based on the severity of symptoms.

From these two heatmaps, gastrointestinal contributes to the topic 1, whereas sleep impairment and psychiatric contribute to the topic 2 for both types of patients. These two graphs also illustrate the differences between well and unwell patients in the severity of symptoms and their distributions in different topics. For example, headache, neuropathy, and twitching dominate topic 3 and topic 4 for unwell patients, whereas memory loss and cognitive impairment account for topic 4 for well patients. We can interpret more information with knowledge of Lyme disease to determine the relationship between different severity levels of symptoms to the status of a patient using these two maps.

7.2. Semi-supervised NMF on Neurological Symptoms. We first tested our Semi-supervised NMF algorithm on the definition of neurological patients based on symptoms. Patients with at least two neurological symptoms rated higher than moderate are labeled as neurological. Neurological symptoms include cognitive impairment, neuropathy, sleep impairment, headache, memory loss, and twitching. The extent to which a symptom contributes to the label is signified by its color, with the brightest representing the greatest and the darkest representing the least. And we achieved an F1 score of about 95.09% and an accuracy of about 91.38% for predicting test labels set after training and validation, proving our algorithm on Lyme data is informative.

Then we compute SSNMF following preprocessing steps mentioned in section 6.2 to study which features contribute to neurological manifestations for unwell chronic patients. As shown in the four heatmaps below, we tested our algorithms several times and achieved different F1 scores and accuracy depending on the importance of the selected features. From left to right, we can observe an increase in both F1 score and accuracy which indicates improved model performance with balanced classes and its ability to classify patients correctly. We can see without the contribution of recalling a tick bite and the length of time noticed after a tick bite, the SSNMF performs comparatively poorly with the lowest accuracy 57.18% among the results in classification. And by including whether a patient recalls a tick bite and the length of time for a patient to notice the tick bite, the accuracy improved to 60.96%. Therefore, this improvement shows these two features might help determine the neurological label for Lyme patients. In the four heatmaps, we can observe a reduced number of tick-born

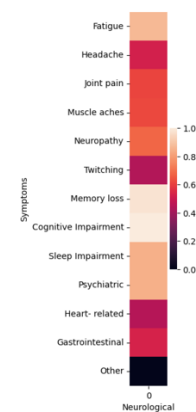


Figure 17 SSNMF: heatmap of BA (features by label) using 12 symptoms for predicting neurological label on unwell chronic patients

co-infections contributing to predicting the label while increasing accuracy. A reasonable explanation is that specific co-infections are not the dominating factors for this classification task. However, whether a patient has a tick-born co-infection is still necessary for classification as shown in the last heatmap where we achieved the highest accuracy of 73.80%. And the period of time for diagnosis and misdiagnosis is always shown as important indicators for determining neurological patients, having the brightest color among all heatmaps we computed. Therefore, we can conclude that five features are important in determining whether a patient is neurological: whether a patient recalls a tick bite, the length of time for a patient to notice a tick bite, the period of time for diagnosis, misdiagnosis, and tick-born co-infections.

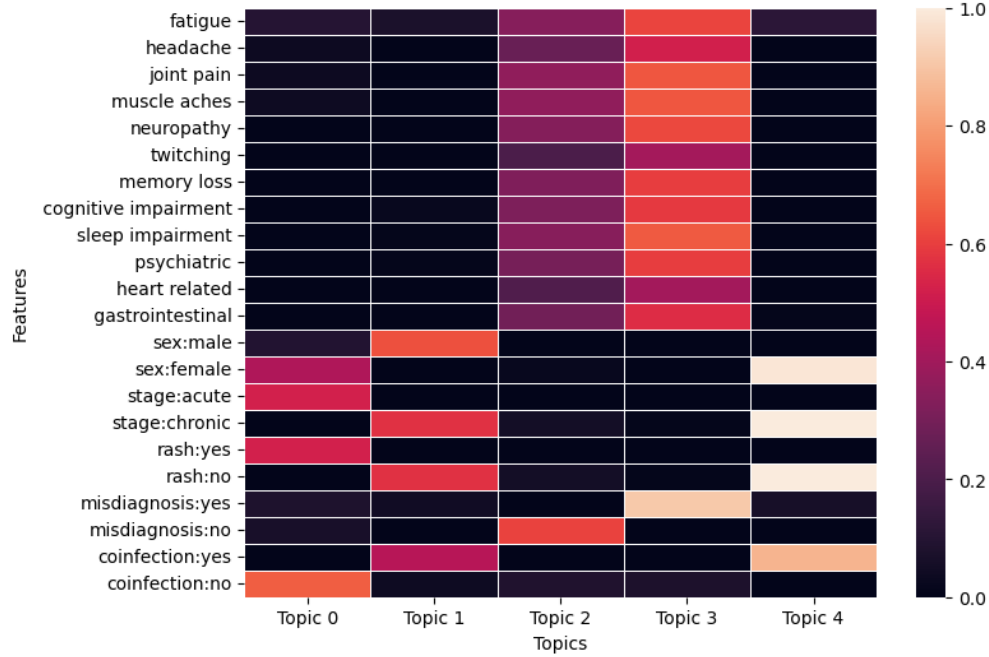


Figure 18 SSNMF: Four heatmaps of the product of BA (features by label) with accuracy and F1 scores for neurological label on unwell chronic patients using selected features for determining informative features in classification

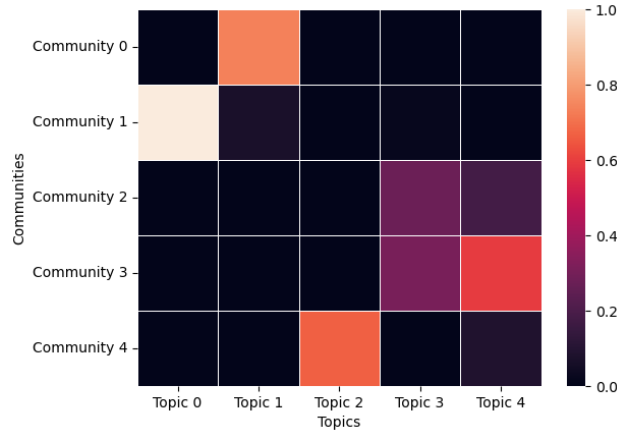
7.3. NMF-Based Community Detection on Patient Responses. We based this experiment on the pre-processed MyLymeData as elaborated in Section 6. In constructing a Lyme patient network suitable for applying the algorithm while maintaining consistency, we made some additional adaptations. We filtered out non-US responses and extracted a set containing all the unwell patients who have answered the symptom-related questions and provided explicit information (excluding "not know" answers) for their basic conditions including sex, co-infection, prior misdiagnosis, etc. This gave us a subset containing 2123 patients ($n = 2123$).

We then approached the problem by experimenting on a 2-layer structure ($N = 2$). Edges in the first layer represent the similarity between patients evaluated by comparing the severity scores of 12 symptoms that are self-assessed by patients. Edges in the second layer are established by comparing the concatenations of 5 basic conditions of patients, namely sex, stage of disease, presence of rash, presence of coinfections, and whether they received a prior misdiagnosis. These 5 features take in binary information and are one-hot encoded.

The resulting community indicator $\mathbf{H}$ is derived by setting the number of communities to be 5 ($k = 5$). The community clustering results contained in $\mathbf{H}$ are then used in establishing the binary class label matrix $\mathbf{Y} \in \mathbb{R}_+^{n \times k}$ following the rules explained in the context of Equation 3.2. An inspection of feature contribution to the clustering of the 5 patient communities through running SSNMF is shown in Figure 19. The accuracy scores of SSNMF compared with community detection results are as the following: homogeneity= 0.57101, NMI= 0.57070, RI= 0.80662.



(a) Heatmap of **A** matrix (features by latent topics)



(b) Heatmap of **B** matrix (communities by latent topics)

Figure 19 With reference to Equation 3.2, we run SSNMF with 20,000 iterations and $\lambda = 0.2$ and derive the **A**, **B** matrices of SSNMF. Figure 19a is the heatmap for **A** that plots feature contribution to the latent topics identified by SSNMF. Figure 19b is the heatmap for **B** that plots the degree of emphasis of the latent topics in each detected community.

Figure 19 displays some interesting patterns of how features are weighed in latent topics. For example, the plotting of Topic 2 and Topic 3 in 19a shows a similar consideration of symptom severity scores indicated by a similar layout of bright colors, while they differ mainly in their emphasis on the presence of prior misdiagnosis. Meanwhile, Figure 19b also presents some degree of interpretability, in which each latent topic demonstrates its particular importance in the formation of one of the communities. In potential future work, we can vary the number of communities and topics we want to detect and experiment on different compositions of features and layers. Overall, generating clusters in the patient network

aided by an SSNMF evaluation helps us identify the shared characteristics in a complex structure and, in a practical aspect, may make specialized help for patients more accessible.

7.4. Classification. We performed classification tasks to predict well/unwell, early/late stage, and before/after 1 year to accurate diagnosis on patients from MyLymeData.

7.4.1. Features Used in the Classification. From the MyLymeData survey, we selected sex, ethnicity, income, period of time to accurate diagnosis, race, prior diagnosis before Lyme, co-infections, and severity ratings of various symptoms as features in our data. Aside from the symptoms question, all participants received the other selected feature questions as part of the baseline survey. The symptoms questions were concatenated from two different questions as a result of survey branching: participants who had answered their current status was unwell in a previous question ranked the current severity of each symptom. Patients who answered their current status were well rated the severity of each symptom when their Lyme disease was at its worst. The ratings for well/unwell were combined into a question ranking severity of each symptom.

7.4.2. Well vs. Unwell Classification. To perform classification on well/unwell patients, the target variable was the current status of participants with Lyme disease. Participants were either labeled unwell (sick, or not well) or labeled well (well in remission, and not on treatment). We perform classification using Naive Bayes, Logistic Regression, SVM, and Random Forests.

We evaluated the models on Accuracy, F1 scores, and demographic parity. After running classification on well/unwell, we used various techniques to determine fairness for biological sex. Using Fairlearn metrics, we determined demographic parity using sex as a sensitive attribute. Scores for each of the classification models are displayed in Table 2, under the imbalanced columns. While logistic regression and random forest had the highest accuracy rates, the models had poor F1 scores. Due to the imbalanced data, with the majority of participants classified as unwell, both models predicted participants to always be unwell, and the models were not actually accurate in classifying between well/unwell. There were 7796 unwell patients used in the classification model, while only 598 were well patients.

To balance out the data, we restricted our data frame to have an equal number of well and unwell patients. We randomly selected 598 unwell participants and re-ran the classification models using the restricted, 1196 person data frame, with 598 well and unwell patients. The classification was run 10 times, each with a new subsample of 598 unwell patients. Table 2 shows the average model performance between the ten runs with balanced data, compared to the imbalanced data.

	Imbalanced			Balanced		
	Acc	F1	Dem. Par.	Acc	F1	Dem. Par.
Logistic Regression	0.93	0	0	0.67	0.70	0.38
Random Forest	0.92	0	0.01	0.71	0.74	0.35
SVM	0.93	0	0	0.53	0.60	0.21
Naive Bayes	0.54	0.15	0.58	0.65	0.72	0.27

Table 2 Accuracy and F1 scores for Well/Unwell Classification on MyLymeData.

While balancing the data decreased the accuracy and demographic parity for most of the models, we can see that it increased the F1 score significantly. Balancing the data for this classification task was crucial to be able to build a model that learns portions of the data. However, it is important to note this model was not highly accurate, and should not be used in classifying unlabeled unwell/well data.

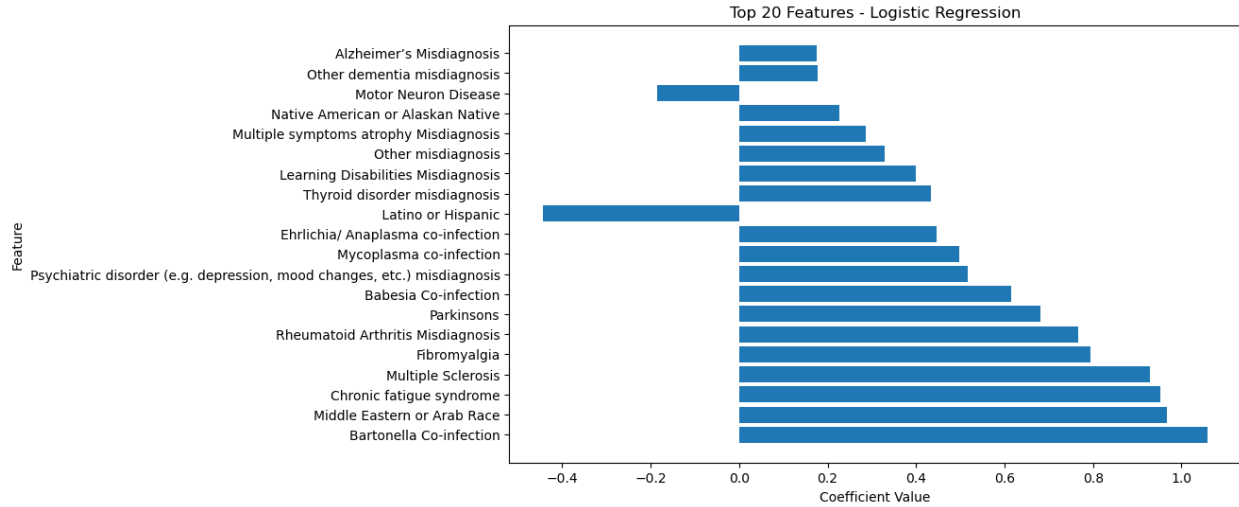


Figure 20 Top 20 Coefficients in Logistic Regression Classification for Early vs. Late Stage Chronic Lyme

7.4.3. *Late Stage/Chronic vs. Early Lyme.* We performed a similar procedure to classify whether someone had Early or Late stage/Chronic Lyme disease. While late-stage/chronic Lyme disease typically is not combined, in order to perform binary classification, we The features were consistent with the well/unwell classification. Both participants who received a tick bite but had no symptoms and participants who did not know their current health status in regards to Lyme were excluded from the analysis. The early label was assigned to participants who answered they had initial Erythema Migrans rash or disseminated Lyme disease (flu, neurological, cardiovascular, or musculoskeletal symptoms). The Late stage/ chronic label was assigned to participants who were either late-stage untreated Lyme disease (6th months after the onset of symptoms) or had chronic Lyme disease (remained ill for at least 6th months after antibiotics treatment).

We performed a similar classification using a Logistic Regression, Random Forest, SVM, and Naive Bayes model. Based on results shown in Table 3, both Logistic Regression and Random forest models performed well, with accuracy and f1 scores above 90 %.

	Accuracy	F1 Score	Demographic Parity	False Negative Rate
Logistic Regression	0.92	0.95	0.01	0.01
Random Forest	0.91	0.95	0.06	0.06
SVM	0.72	0.83	0.41	0.32
Naive Bayes	0.38	0.50	0.31	0.69

Table 3 Accuracy and F1 scores for Early vs. Late Stage/Chronic Classification on MyLymeData.

Because the Logistic Regression performed had the highest accuracy, f1 score, and lowest demographic parity and false negative rate, we did a further analysis of the logistic regression model to determine the features with the largest coefficients informing the model. Table 4 depicts the top 5 coefficients in magnitude in the model, with Bartonella Co-infection and Middle Eastern or Arab race option features scoring the highest. Additionally, Figure 20 shows the top 20 coefficients in magnitude for the logistic regression. We can interpret the coefficients of each feature as a change by one unit in the feature changes the odds ratio of being classified by a factor of the coefficient exponentiated.

	Coefficient
Bartonella Co-infection	1.0588
Middle Eastern or Arab Race	0.9664
Chronic fatigue syndrome	0.9521
Multiple Sclerosis	0.9287
Fibromyalgia	0.7936

Table 4 Top 5 Feature Coefficients in Logistic Regression Classification for Early vs. Chronic/Late stage Lyme Disease.

	Coefficient
Psychiatric disorder	0.8670
Alzheimer's	0.8323
Chronic fatigue syndrome	0.7890
Thyroid disorder	0.7863
Fibromyalgia	0.7190

Table 5 Top 5 Feature Coefficients in Logistic Regression Classification in less than 1 year vs. over 1 year after symptom onset before accurate diagnosis

Further inspection of Bartonella Co-infection revealed that about 15 % of those with no co-infection had early EM, however, of those with a Bartonella Co-infection, only 2 % had early EM. Similarly, analysis on the second largest coefficient informing the logistic regression, Middle Eastern or Arab race. Interestingly, none of the participants who identified as Middle Eastern or Arab had Early Lyme disease, compared to 15 % of participants who were not Middle Eastern/Arab.

7.4.4. Classification on Time to Accurate Diagnosis. We performed another classification task to classify between those who were accurately diagnosed within a year of symptom onset, compared to those who took more than a year after symptom onset to be accurately diagnosed. The features used in the code were sex, ethnicity, income, race, prior diagnosis before Lyme, co-infections, and severity ratings of various symptoms, consistent with both the well/unwell and early/chronic classifications. The Logistic Regression model had the highest performance in both accuracy (74% accurate) and in f1 score (%78). Extracting the features with the highest coefficients, depicted in table 5, we found that various prior misdiagnoses were the most influential in predicting a diagnosis that took over one year.

Following Logistic Regression to classify time to accurate diagnosis, we compared the results of the model to a supervised NMF model. We ran SSNMF as noted in section 4.2, with labels less than a year vs. over a year as our label matrix. The data matrix was consistent with other classification techniques, using sex, ethnicity, income, race, prior diagnosis before Lyme, co-infections, and severity ratings of various symptoms. Because we did not finish the SSNMF analysis, future work is needed to specify a λ value with a validation set and record the model's accuracy.

We first analyzed Matrix **A** and **B** through heatmaps. **A** depicts the weights of questions by topics, and **B** depicts the weight of each topic for classification of time to diagnosis??. The rows of Matrix **A** represent the questions used in the factorization, although are left out of the heatmap for visual purposes. The relative reconstruction error was 0.24, and, because the data was normalized, this error ranged from 0 to 1, and was fairly low for our matrix.

We multiplied the Question by Topic matrix by the topic importance matrix and determined which questions were most important in the classification of time to diagnosis before vs. after 1 year. Figure 21 denotes a heatmap of all question importance, while Table 6 shows the top 10 questions with their corresponding coefficients.

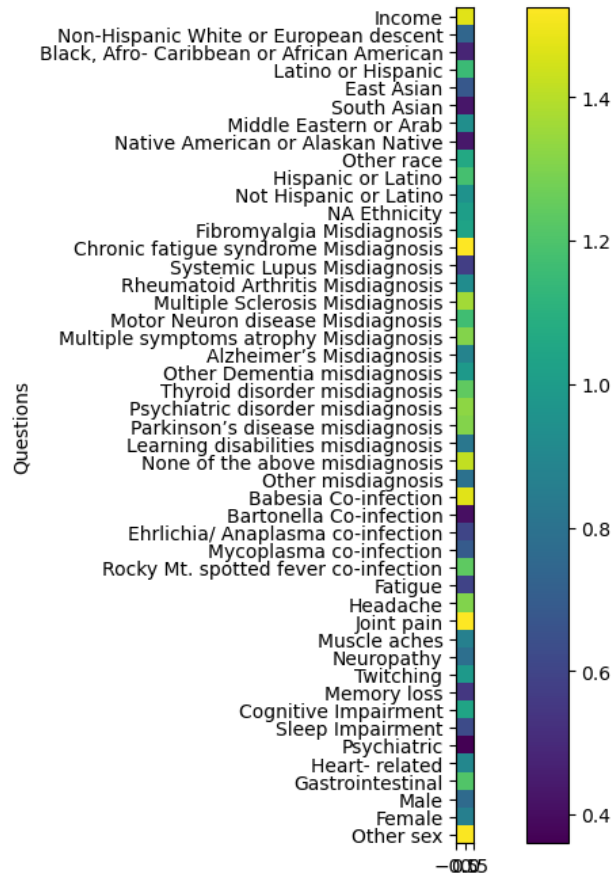


Figure 21 Question weights in SNMF classification of time to an accurate diagnosis.

	Coefficient
Twitching symptom	0.99
NA Ethnicity	0.97
Rocky Mt. spotted fever co-infection	0.95
Native American or Alaskan Native	0.94
Learning disabilities misdiagnosis	0.94
East Asian Race	0.93
Multiple symptoms atrophy misdiagnosis	0.93
Fibromyalgia misdiagnosis	0.89
Other Dementia misdiagnosis	0.89
Latino or Hispanic Race	0.88

Table 6 Top 10 questions in SNMF classification of time to accurate diagnosis.

The question weight matrix plays a significant role in understanding the importance of each question when classifying whether an individual was diagnosed with Lyme disease before or after one year. The elements represent the strength of association between questions and topics, weighted by the importance of each topic in the classification task. Higher coefficients suggest that a particular question is more influential in distinguishing between individuals diagnosed with Lyme disease before and after one year. In our data, we can see that twitching symptom, NA ethnicity, and Rocky Mt. spotted fever co-infection

were the most influential features in our model to classifying before/after a year. Of course, further analysis is needed to understand these coefficients, namely, optimizing the λ value, and determining the accuracy of the model is necessary before interpretation of the coefficients.

7.5. Fair NMF. We perform preliminary FNMF on MyLymeData 23. To compare it with Vanilla NMF, observe that the losses for three subgroups converge at rank 12, and the dataset we are using has 12 features. With Preliminary FNMF, the subgroup losses achieve early convergence at around rank 9.

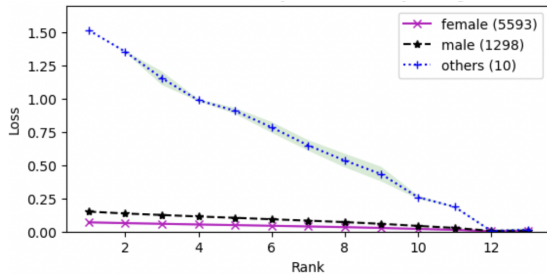


Figure 22 Vanilla NMF on unwell Lyme patients grouped by biological sex from symptom dataset containing 12 features

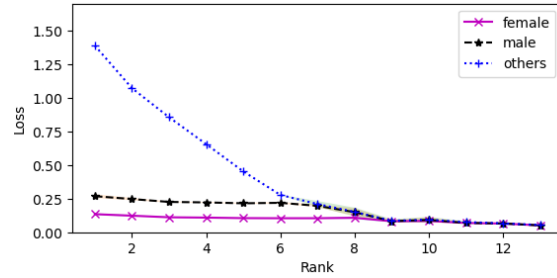


Figure 23 Preliminary FNMF on unwell Lyme patients grouped by biological sex from symptom dataset containing 12 features

We look closer at the effects of preliminary FNMF and reweighted FNMF, examining whether reweighting the dictionary matrix is indeed helping mitigate the fluctuations of losses between groups. For the graphs above (changed to log-scale for visualization) 24, preliminary FNMF and reweighted FNMF achieve a similar effect by having a convergence of losses at around rank 9. However, something to note is that the descent for reweighted FNMF is smoother and more stable, and the curves also converge with smaller gaps between different subgroups.

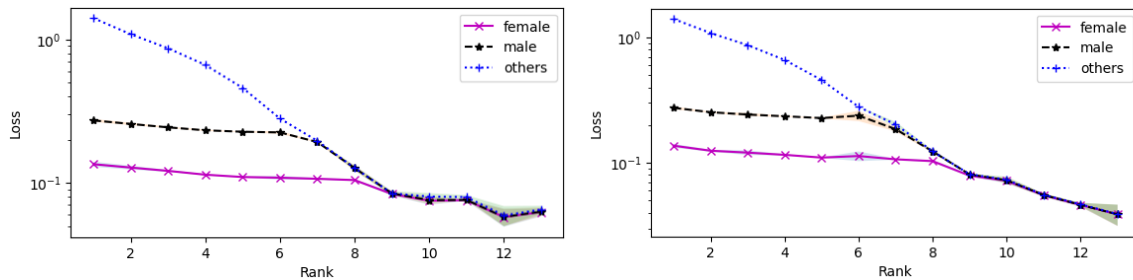


Figure 24 Left: Preliminary FNMF on unwell Lyme patients grouped by biological sex (log-scale); right: reweighted FNMF on unwell Lyme patients grouped by biological sex (log-scale)

Next, we look at a zoomed-in graph on the last 20 iterations. For comparison, the scale of the y-axis for the reweighted FNMF graph shown on the right is changed to match the scale of the preliminary FNMF. Fluctuations are tremendously improved for reweighted FNMF, resulting in a much lower loss.

7.6. Anomaly Detection. In this section, we focus on the sub dataset of unwell patients and their symptoms to determine if there were any patient anomalies with preprocessing outlined in Section 6.2. First, we run iForest 100 times, due to randomness, using a contamination factor of 0.5 to determine the anomalies that appeared 70% of the time, which were then listed as our consistent anomalies. Through this

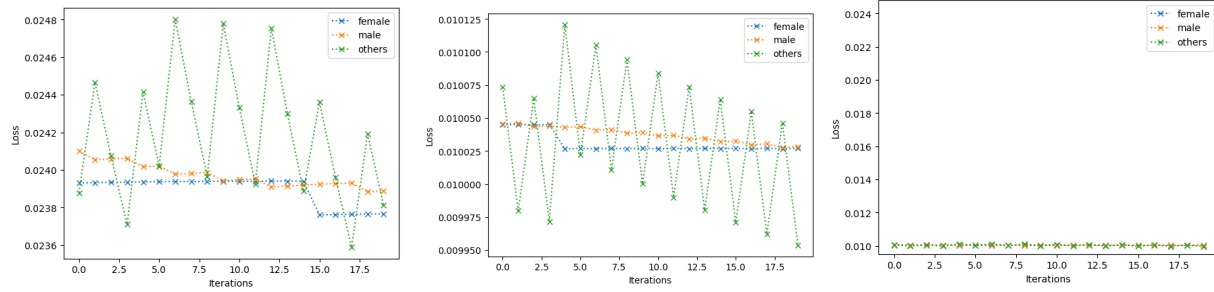


Figure 25 Left: fluctuation of loss for preliminary FNMF on unwell Lyme patients grouped by biological sex (last 20 iterations); middle: the fluctuation of loss for reweighted FNMF on unwell Lyme patients grouped by biological sex (last 20 iterations); right: the fluctuation of loss for reweighted FNMF with changed scale (last 20 iterations)

algorithm, 3.9% of unwell patients were determined to be anomalies. We also run LOF using a contamination factor of 0.05. LOF is not rooted in randomness, meaning it was run only once to determine the anomalies. Through this algorithm, 5.01% of unwell patients were determined to be anomalies. By implementing both of the anomaly detection algorithms, iForest and Local Outlier Factor, we were able to determine the anomaly patients. Between the two anomaly detection algorithms, the percentage of overlap is 2.6%, this is 0.23% of total unwell patients.

We applied SSNMF 3.3 with three topics to study the detected anomalies from both algorithms. We extract the topics or patterns that are shared between the detected anomalies and non-anomalies and the topics that distinguish between them.

The data matrix $\mathbf{X}$ was represented by the collection of unwell patients and their symptoms. While the label matrix $\mathbf{Y}$ was represented by a one-dimensional array, with 1 representing that the patient is an anomaly and 0 otherwise. To ascertain the optimal value of λ that would yield the greatest F1 score, an experimentation phase was conducted where various values from 0 to 1000 in intervals of 0.01 of λ were tested. A table 7 illustrating the results of this experiment is shown below.

Best Lambda for SSNMF			
Algorithm	Lambda	F1	Accuracy
iForest	202.2	0.1	0.68
LOF	131.31	0.16	0.75
Combined	6.06	0.87	0.77

Table 7 Comparison of different λ values for SSNMF for each anomaly algorithm.

As shown in the provided Table 7, both iForest and LOF demonstrate a relatively low F1 score. This outcome can be partly attributed to an imbalanced dataset that features a limited percentage of anomalies of 3.9% and 5.01% respectively. Nonetheless, upon combining the anomalies detected by both algorithms, the anomaly proportion nearly doubled to 8.8%. As a result, the F1 score and accuracy of the model increased significantly.

Below are the results of SSNMF from using iForest, LOF, and a combination of the anomalies detected from both algorithms.

Based on the heatmaps in Figure 26a, for iForest, the symptoms muscle aches and headache carry the greatest importance in the prediction of being an anomaly. Whereas for LOF in Figure 26b it is fatigue and heart-related symptoms. However, combining both sets of anomalies yielded a significantly higher F1 score and accuracy, thus it would be more beneficial to make interpretations based off of the combined

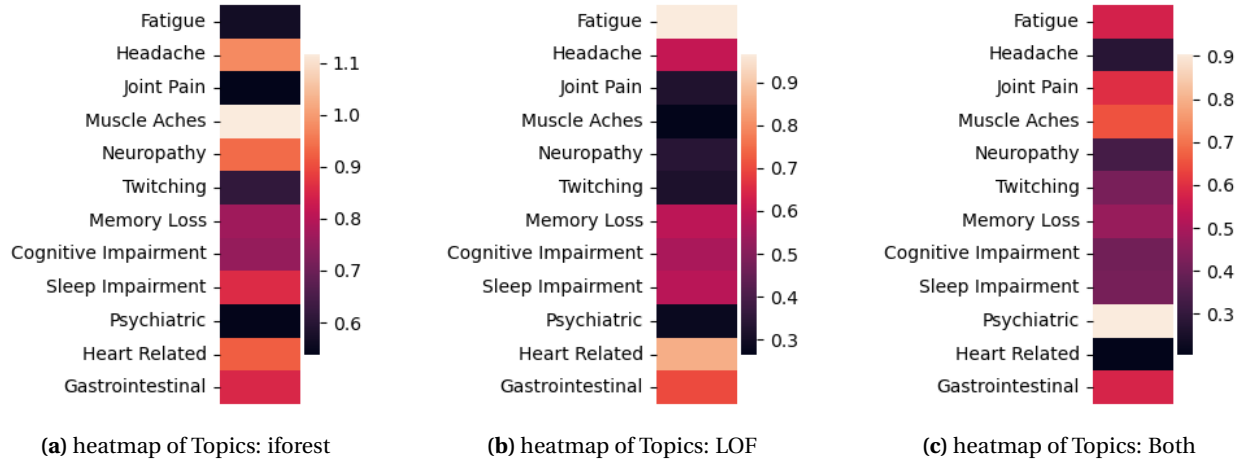


Figure 26 Results of SSNMF on the different set of anomalies from each detection algorithm that show the symptoms of greatest importance when classifying a anomaly.

set of anomalies. Therefore, when combining the anomalies found from both algorithms and running SSNMF 26c – psychiatric and muscle ache symptoms hold the greatest significance.

8. CONCLUSION

In this report, with an objective to explore trends and reveal insights in MyLymeData, we use various machine learning techniques to fully explore possible research topics in Lyme Disease treatment. Non-negative matrix factorization serves as our main framework, where we also extend its purpose to different areas and bridge the technique with many other existing methods. First, we discovered important symptoms contributing to patients' status and indicators for determining neurological patients with unsupervised and supervised NMF. As well as used various classification techniques to predict patient status through a collection of patient responses. After this, we determined symptoms that have a greater impact on determining whether a patient is an anomaly via supervised NMF. Another aspect we explored was identifying communities in a multiplex network structure with decent interpretability. We also detected clusters using feature similarity and performed clustering based on rough splits given by Neural NMF. Then, by rescaling the dataset, we observed a non-linear relation between loss and the scalars. After this, we proposed a Fair NMF algorithm where different subgroups converge on lower rank compared to vanilla NMF and result in a fairer low dimensional representation. And concluded that early stopping does drastically shrink the number of iterations needed but yields a higher loss than the proposed Fair NMF algorithm. Below, we provide insight into possible future directions for each of our various topics

8.1. Unsupervised and supervised NMF on symptoms and neurological manifestations. Future directions of unsupervised NMF will involve features such as treatments and other potential conditions that lead to well and unwell status for Lyme patients. For classifying neurological patients with supervised NMF, we may include more features to study their contributions to this label and examine how we can improve the accuracy of label prediction.

8.2. Classification. Future directions of classification include fine-tuning the models to increase accuracy and F1 scores. Additionally, incorporating fairness constraints into the objective functions would be beneficial in increasing fairness. Finally, exploring various other classification techniques and incorporating different features to determine tasks that best predict various Lyme disease labels.

8.3. Anomaly Detection. Future directions of anomaly detection will be to shift from amalgamating sets of anomalies to the adjustment of anomaly-to-not anomaly instances ratio. This adjustment will be achieved by introducing a controlled random selection process that targets a specific proportion of non-anomalies. This strategic alteration will effectively enable us to finely tune and balance the ratio between anomalies and non-anomalies within the dataset, thereby enhancing its overall equilibrium and increasing the F1 score.

8.4. improvements on Neural NMF via feature similarity-based cluster detection and clustering. Future steps for feature similarity-based cluster detection and clustering could be to expand the use of such algorithms to data that is not necessarily tree-structured. Real tree-structured data is limited, and most of the data available for investigation are not hot encoded. It would also be interesting to further investigate the range of quantile threshold α that would guarantee reasonable results.

8.5. NMF-Based Multiplex Community Detection. Future works in this direction can include a more comprehensive experiment on MyLymeData using varying parameters for communities and latent topics. If seeking further advancement, we may also develop a framework for determining the trade-off parameters to support the algorithm with more concrete reasoning.

8.6. Fair NMF. With established fair NMF algorithms in this paper, we are able to achieve a "fairer" lower dimension representation for a given dataset that contains different subgroups based on certain attributes. This result has many implications in the field of social science research that requires an equal representation of different subgroups. In reality, datasets that under-represent minority groups can make use of our fair NMF as a preprocessing step to achieve fairness.

There is a lot of potential for improving the proposed fair NMF algorithm in the future. In a broader sense, one could implement a rescaling method for n populations (where $n > 3$) and may use a data-driven or mathematical approach to find the optimal values for 4 or more unknown constants c_i . To further boost the robustness of the algorithm, we can develop more helper functions to optimize for parameters and do hyperparameter tuning on learning rate η . In specific, weights on the dictionary matrices of which we are keeping track can be determined with more technical methods like grid search. Another future direction is to fuse the reweighted FNMF algorithm with an early stopping strategy together to achieve efficiency.

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